

**Report on the
Migration of Birds
in the Autumn of 1879**

Harvie-Brown and Cordeaux



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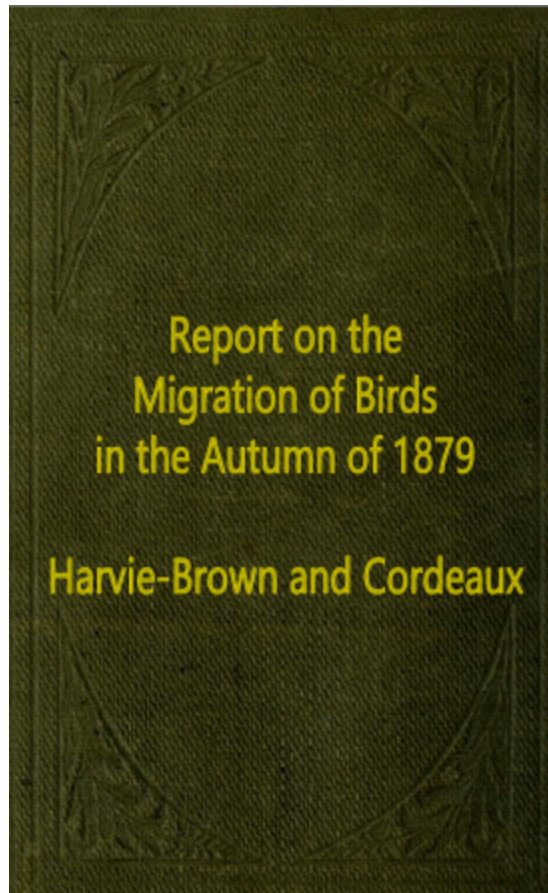
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**REPORT ON THE
MIGRATION OF BIRDS IN THE
AUTUMN OF 1879.[\[1\]](#)**

By John A. Harvie-Brown and John Cordeaux.

[1] I would here direct attention to another paper of mine bearing upon this subject, which was read at the meeting of the Glasgow Natural History Society on Sept. 30th, 1879, and forms part of its 'Proceedings' for the Session 1879-80, now in the press. It forms a Report upon Migration of 1878, Journal of the severe winter of 1879-80, and Observations on the effects of the weather, under the different species of Mammals and Birds noticed. I read a similar Report on 1879-80 at the meeting of the same Society in March last.—*J. A. H.-B.*

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EAST COAST OF SCOTLAND.

Printed forms of enquiry and letters of instructions were sent to twenty-six lighthouses on the east coast of Scotland. Thirteen stations have sent in returns, and thirteen have either sent in none, or have returned blank forms, owing to unusual scarcity of birds. The stations from which co-operation was asked are the following, commencing with the most northerly. Those from which returns have been received are marked with a *.

*North Unst, Shetland; white and red sectors, fixed. Robert Burnett.

Whalsey Skerries, Shetland; white, revolving every minute.

Bressay Sound, Shetland; red and white alternately, rev. every minute.

*Sumburgh Head, Shetland; white, fixed. William Anderson.

*North Ronaldshay, Orkney; white, flash every 10 seconds. John Tulloch.

Start Point, Orkney; red, fixed.

*Auskerry, Orkney; white, fixed. Charles C. Irvine.

*Hoy Sound (Low), Orkney; white, fixed. Alex. Harp.

" (High) Orkney; red and white sectors, fixed. W. Gordon.

Cantick Head, Orkney; white, revolving every minute.

*Pentland Skerries, Orkney; white, fixed. D. M'Donald.

*Dunnet Head, Caithness; white, fixed. George M'Lachlan.

Holborn Head, Caithness; white and red, flash every 10 sec. D. Charleson.

*Noss Head, Caithness; white and red, revolving every $\frac{1}{2}$ m. Alex. Creig.

*Tarbet Ness, E. Ross-shire; white, int. visible every $2\frac{1}{2}$ m. W. Davidson.

Cromarty, E. Cromarty; red, fixed. Robert S. Ritson.

Chanonry Point, Elgin; white and red sectors, revolving every minute.

Covesea Skerries, Elgin; white and red sectors, revolving every minute.

- Kinnaird Head, Aberdeen; white and red sectors, fixed. F. Harvey.
Buchan Ness, Aberdeen; white, flash every 5 seconds. Thomas Gallie.
*Girdleness, Aberdeen; white, fixed. Wm. Gulcher.
Montroseness, Forfar; white, fixed. Patrick E. Reid.
*Bell Rock, off Coast of Fife; white and red revolving. James Jack.
*Isle of May, Firth of Forth; white, fixed. Joseph Agnew.
*Inch Keith, Firth of Forth; white, revolving every minute. R. Grierson.
St. Abbs Head, Berwick; white, flash every 10 seconds. Robert Seater.

A general scarcity of birds is reported by a great majority of the observers on our east coast, which partly accounts for the absence of returns from several stations. Those sent in have been carefully prepared by the various reporters so far as materials allowed. Excluding Swans, Geese, Ducks, and Rock-birds, notes have been made on about twenty-eight different species.

SEPARATE REPORT OF EACH SPECIES OBSERVED.

At North Ronaldshay, an Owl—species unknown, but probably the Short-eared Owl, *Asio accipitrinus*—was seen on Nov. 3rd, flying south, at 10 a.m.; and at Dunnet Head a Horned Owl, also probably of the same species, struck three times, but got away. This was at 6 p.m.

From Pentland Skerries we have all the records of hawks. Two early records (July 7th and 14th) may refer to autumn migration or not. Thus, "a black hawk" was seen on July 7th at 3 p.m., and on the 14th another of the same at 7 p.m. On Sept. 17th one hawk struck at 11.30 p.m. The species is probably, the Sparrowhawk, *Accipiter nisus*.

Accounts forwarded by other correspondents, however, show that Hobbies (*Falco subbuteo*, Linn.), visited the Lewis in some numbers this year.^[2] At Sumburgli Head the appearance of a White-tailed Eagle is recorded, but is only of local interest, as these birds are known to breed near that station.

That a migration, however, does take place amongst Eagles, and mostly of this species, is undoubted. Mr. D. Dewar, an experienced ornithologist, writes me:—"Every winter, for this longtime back, when we have a strong east wind in November, Eagles and Rough-legged Buzzards appear over Loch Tay. In November, this year, I saw four Eagles together, all White-tailed, and three Rough-legged Buzzards along with them."

[2] I have referred more fully to this in my "Report on Scottish Ornithology" for the Nat. Hist. Soc. of Glasgow for 1879-80.—J. A. H.-B.

SWIFT, *Cypselus apus*.—Only one at Sumburgh Head on May 25th, doubtless on the spring migration. Two were seen at 10 a.m., wind light S.W., breeze and haze. "Swifts, however, do not breed in Shetland" (*vide* Saxby's 'Birds of Shetland,' p. 147).

SWALLOW, *Hirundo rustica*.—I have returns from four stations, Sumburgh Head, Aukerry, Tarbet Ness, and Isle of May. At the northernmost station (Sumburgh Head) "a number were seen flying about," at 11 a.m., wind at W.S.W., fresh and hazy. But this record was on June 12th, so probably refers to the spring migration. Their earliest appearance noted is August 21st, when four struck at Aukerry at 1 a.m., wind S.E., strong with fog; the latest Oct. 27th, when two were killed at Tarbet Ness at 7.10 p.m., wind N.W., light with fog. This shows migration between Aug. 21st and Oct. 27th. A "rush" appears to have taken place at the Isle of May on Sept. 19th, at 2 a.m., wind light E. with fog, when a flock of young birds was seen and four of them were wounded against the glass. Swallows migrate in August and September with an E. to S.E. wind, light to strong. In October my only record at Tarbet Ness gives wind N.W., light with fog. When the wind has been N.W., S.E. or E. there has been fog; when, in summer, at Sumburgh Head, W.S.W., there has been haze and rain. The last Swallow was seen in Berwickshire by Mr. J. Hardy on the 30th Sept. 1879.

MARTIN, *Hirundo urbica*.—From Pentland Skerries one record in which this species is distinguished from the Swallow. On August 25th one struck at 2.30 p.m. during a light W. wind with haze. The last was seen in Berwickshire by Mr. J. Hardy on the 3rd October, 1879.

WREN, *Troglodytes europæus*.—At North Ronaldshay five seen during the daytime on Nov. 21st, fresh breeze (direction not noted) with haze.

Reported as usually appearing in large numbers at Auskerry, but the above-mentioned were the only ones seen.

GOLDCREST, *Regulus cristatus*.—The almost total absence of this species is specially remarked at the Isle of May. Usually they are abundant. These birds were decidedly scarce in Scotland throughout the summer.

ROBIN, *Erythaca rubecula*.—Only two records of this species on the east coast. At Tarbet Ness one was killed on August 6th at 10 p.m., wind light E. breeze, fog and rain; and at Inch Keith one struck on August 10th at 11.30 p.m., in similar weather, but with wind light W.

WHEATEAR, *Saxicola ænanthe*.—The most northerly station of the two recorded is Pentland Skerries, and to the south the Bell Rock. Between Sept. 7th—when great numbers struck and were lost at the latter station, and birds continued passing or flying round lantern between midnight and dawn, wind S.E. and variable, fog and haze—and Sept. 17th, Wheatears or "Stonechats" were on passage. If any rush took place it was between Sept. 7th and 11th, on which latter day fourteen were seen and two killed. Winds veered from S. and S.E. on Sept. 9th and 7th, to W. and S.W. on the 11th, 13th and 17th. All the birds passed at night between 8 p.m. and dawn.

TITMICE.—At only one station were Titmice observed on our east coast, viz., at the Isle of May, where some were seen on Sept. 15th at 10 a.m., wind light W. with haze. From Auskerry, however, there is the general report that they are in most years abundant there, though none have been seen this season.

SONG THRUSH, *Turdus musicus*.—Recorded from a number of stations from North Ronaldshay in the north to Auskerry, Pentland Skerries, Dunnet Head, Noss Head, Girdleness, and Bell Rock in the south. The greatest numbers noted were at Pentland Skerries, but great numbers also were noted at North Ronaldshay and the Bell Rock, where many "struck and were lost." Their earliest appearance recorded is on Sept. 16th, when four struck at Noss Head between 8 p.m. and 10 p.m., wind S., moderate with haze. The latest date is Oct. 25th, when numbers, along with Blackbirds, struck at North Ronaldshay. The time thus occupied was from Sept. 16th to Oct. 25th. A rush took place during October, and the greatest rush, perhaps, between Oct. 14th and 25th. At North Ronaldshay, on Oct. 1st, numbers struck, along with Blackbirds, and were flying about all night. Others seen at Girdleness, the Bell Rock, at Pentland Skerries, and Dunnet Head.

Thrushes migrated when the wind was between E. through S.E., S.S.W. to N.W., but principally when southerly. Fog, haze, and rain on all the dates given. Time of migration, after dark and before dawn at all dates.

BLACKBIRD, *Turdus merula*.—Appeared at North Ronaldshay and the Bell Rock between Oct. 1st and Nov. 25th. A rush during this time recorded on three dates:—Oct. 1st, "Numbers flying about all night," at North Ronaldshay, wind S.E., strong, "along with Thrushes." Oct. 14th, four (two males and two females) killed; number flying about between 8 a.m. and dawn; wind N.E., fresh, with haze and rain. Nov. 25th, numbers, along with Thrushes, flying about all night; wind light E. breeze, haze.

RING OUZEL (or "Mountain Blackbird"), *Turdus torquatus*.—Reported from Sumburgh Head only, where one struck at 4 a.m.; wind light N. at 5 a.m., S.W. gale at 4 p.m.; weather clear. A number of the same species were seen in the country about the same time, and supposed to be young birds. At Aukerry, where they generally pass in large numbers, none were observed.

Besides the above, "large Thrushes," probably Fieldfares,^[3] passed or were caught at Aukerry, in October; four on the 15th between 1 and 4 a.m., wind light N., clear; six on the 16th between 2 and 4 a.m., wind S.S.W., strong, haze; and two on the 22nd at 3 a.m., wind strong S.W., and haze. Large numbers struck the lantern at the Bell Rock, and were lost in the sea between midnight and dawn on Sept. 7th, and again at the same station on Oct. 14th, between 3 a.m. and dawn. These were mixed in flocks of Blackbirds, Thrushes, and "various" other species.

^[3] Afterwards identified as such by the head, wings and tail sent in later communication.

The great scarcity of Thrushes in Scotland all summer was caused by the severe winter of 1878-79 and late spring of 1879, a consequent "crowding down" upon lower latitudes taking place during that breeding season. I have referred very fully to this in my first Report on Scottish Ornithology for the Natural History Society of Glasgow for 1878-79 (now in MS., ready for press), read Sept. 30th, 1879. This scarcity is still felt. In the birch-woods of the west coast I saw small flocks on Oct. 28th, 1879.

ROOK, *Corvus frugilegus*.—At Sumburgh Head numbers were seen about 9 am on Nov. 2nd, wind N.N.W., strong to fresh, with sleet and snow showers. "Rooks do not breed in Shetland, but are often seen in the spring

time, but very seldom in the fall, crossing, as is supposed, between Norway and Scotland." At Dunnet Head about 100 Rooks, mostly young birds, were seen at 12 noon, with fresh breeze from S.E. to S.W.

STARLING, *Sturnus vulgaris*.—Noted at only two stations. At Pentland Skerries on October 19th six struck between 6 p.m. and 4 a.m., wind S.W. and rain; on the 26th one at 8 p.m., wind W., and clear; and on Nov. 20th two at 6 a.m., wind S.W., strong and haze. At Dunnet Head, on Oct. 20th, seven struck and four were killed at night, wind N.W. and N.E., haze.

SISKIN, *Carduelis spinus*.—At Inch Keith three young birds struck at 1 a.m. on Sept. 16th, wind light W., fog.

LARK, *Alauda arvensis*.—Pretty generally recorded, but scarcer than usual; the northernmost station Pentland Skerries. They are also noted at Tarbet Ness and Girdleness. The most counted was thirty-six at Pentland Skerries between 5 p.m. and 4 a.m. on October 13th-14th, and the next largest number was sixteen, on Sept. 15th, between 8 and 10 p.m. Their earliest appearance was on August 16th, when four struck at midnight at Tarbet Ness, wind light S.E. and haze; the latest Nov. 19th, when nine struck (four killed) at the same station at 8.10 p.m., wind S.W., light and haze. The time occupied is thus between August 16th and Nov. 19th. A rush took place between Oct. 13th and 14th, or say about the middle of October, and again about November 19th at Pentland Skerries, Tarbet Ness, and Girdleness.

Larks migrated or came most under notice in S.W. winds, but also in S.E., N.E. (as at Pentland Skerries, when thirty-six were seen), and W. Our data from the Scotch coasts are too limited this year to admit of conclusions, unaided by the English returns. Haze and fog are recorded on all the days noticed. All records give hours of passing at night from 6 p.m. in October, and from 8 10 p.m. in November till 4 or 5 a.m.

WOODPECKER, *Picus (major?)*.—At Inch Keith two struck and an adult bird was killed at 9 p.m. on Sept. 15th, wind W., light breeze and fog.

LAPWING, *Vanellus cristatus*.—Reported to have left the vicinity of Hoy Sound in the third week of September. At Auskerry thirty or forty were seen about 3 p.m. on Sept. 11th, wind light W. and haze, and at Tarbet Ness a large flock was seen at 11 a.m., when the wind was N.W., light and haze on Oct. 12th.

PLOVER, *Charadrius pluvialis*.—At Pentland Skerries a flock was seen at 1.15 a.m. on Sept. 3rd, wind S.W., almost calm, with fog, and another flock at 2 p.m. on Dec. 7th, wind N.W. and clear. These represent earliest and latest dates. At Dunnet Head about 160 Plovers passed with S.E. wind and rain. At Tarbet Ness a large flock seen at 4 p.m., wind N.W., fresh and haze on Oct. 12th. Plovers remained unusually late on the high hills of Perthshire this autumn. On Oct. 15th I saw three on the hills around Glen Queich, and two flocks on the 16th. On the 14th was a snowstorm and ice a quarter of an inch in thickness on hill puddles.

HERON, *Ardea cinerea*.—At Girdleness on October 23rd two Herons were seen at 3.30 p.m., wind S.W., clear.

CURLEW, *Numenius arquata*.—Only three records, one of which no doubt refers to local migration or to late spring or summer. On June 29th a flock was seen at Pentland Skerries at 7.40 p.m. At Sumburgh Head, on July 30th, "a number crying very loud" were heard about 11 a.m., wind W.S.W. (gale on Aug. 1st), with haze and rain. At Pentland Skerries, on Dec. 11th, a flock was noted at 7.40 p.m. during light W. wind and clear. Did the migration last from July 30th to Dec. 11th? Curlews appear to migrate from early dawn to dusk, as far as records show. These birds were nearly a month late in arriving on our coast in Stirlingshire.

SANDPIPER (sp.).—At Pentland Skerries three struck and were killed; wind strong S.W. and haze.

WOODCOCK, *Scolopax rusticola*.—The most northerly station was North Ronaldshay. Thence southward, but nowhere in large numbers, seen or struck at Auskerry, Pentland Skerries, Dunnet Head, Tarbet Ness, Girdleness, and Isle of May. The largest number seen at any station at the same time being five at Tarbet Ness on December 3rd. A single bird is noted at Dunnet Head as early as Sept. 17th. None again till Oct. 15th, when one was killed at North Ronaldshay at 9 p.m., wind N.E., moderate, with sleet. The latest recorded was Dec. 3rd, when, as related above, five were seen at Tarbet Ness at 11.30 a.m., wind strong W., with snow. The length of time occupied in migration was between Sept. 17th and Dec. 3rd. "Rushes" of migrants took place between Oct. 15th and 21st, but the reports are very meagre, being in most cases instanced by single birds striking at North Ronaldshay, Auskerry, Pentland Skerries, and Isle of May. Again a rush

between Nov. 11th and 19th at North Ronaldshay and Girdleness; and again between Dec. 1st and 3rd at North Ronaldshay and Tarbet Ness. In two cases only have more than single birds been recorded. The migration took place principally when the wind was between S. and W.; but I have records of their flights also in strong N. wind with snow, on Dec. 1st, at Tarbet Ness; in N.E., moderate breeze with sleet, on Oct. 15th at North Ronaldshay; and again, with an E. light breeze and clear, on Dec. 21st, at the same station. When the wind has been S.W. or W., there has been fog or haze; when S., rain, as in November, on the 14th and 15th. Woodcocks appear to migrate principally at night, and are observed during the hours between evening dusk and morning grey, or, in other words, between 7.30 p.m. and 5 a.m. Exceptions occur, as at 11 a.m. at North Ronaldshay 11.30 a.m. at Tarbet Ness, 2 p.m. at North Ronaldshay, and 3.30 p.m. at Girdleness; these last being all in daylight. A flight came inland in Stirlingshire about Nov. 22nd, on which day a party of five guns, of which party I was one, killed eighteen in Torwood Covers.

SNIFE, *Gallinago scolopacina*.—At the Isle of May one was seen at 10 a.m. on August 6th, with light S.E. wind, fog and rain. Snipe scarce or much scattered, owing to wet season. Entirely disappeared from inland localities when the hard frost of beginning of December set in, and few returned all winter.

WATER RAIL, *Rallus aquaticus*.—One record at Girdleness; on August 17th, one was seen at 8 p.m., wind E. with rain.

GREY GEESE.—At Dunnet Head, on Sept. 3rd, thirteen Grey Geese were seen "going north" at 3 p.m., wind S.E., fresh, haze and rain.

BERNACLE GOOSE, *Bernicla brenta*.—At Dunnet Head a mixed flock of old and young (twenty-five) seen going W. on Sept. 7th, and five more on the 14th, during the day, wind light W. Fog on the 7th; clear on the 14th. *Anatidæ* were a month earlier in appearing on Loch Tay in the autumn of 1879. Geese were reported as unusually abundant from many localities.

SWANS.—Three Swans seen at North Ronaldshay flying south at 2.30 p.m., with W.N.W. fresh wind and clear, and again at same place at 9.30 a.m., on Nov. 27th, swimming on the water; wind strong N.E., and clear. At Pentland Skerries eight were seen at 2 p.m. on Oct. 26th; wind light W. and clear. At

Dunnet Head three old and one young seen going S.W., at 1 p.m., breeze light, on Sept 14th; and three old birds again on Sept. 22nd.

DUCKS.—At Tarbet Ness a large flock of ducks passed inland at 10 a.m., with a N. gale and snow, on Nov. 12th. At Girdleness, two ducks seen at 4.40 a.m., wind S.W., and haze, on Oct. 16th. At Tarbet Ness a large flock seen at 9.15 a.m., wind W., strong and snow, on Dec. 5th. At Dunnet Head seven ducks (Widgeon) stayed on the lake three days from Oct. 25th; arrived with W. to N.W. breeze and cloudy.

GUILLEMOT, *Uria troile*.—At Pentland Skerries a flock seen on June 25th, at 7.15 p.m.; wind E.N.E., and light haze. (Compare with notes on W. coast at Cape Wrath and elsewhere.)

PUFFIN, *Mormon fratercula*.—At North Unst innumerable Puffins seen on evening of April 24th.

COMMON SKUA (or "Bonxie"), *Stercorarius catarrhactes*.—At North Unst, usually appear about middle of April. "We do not see more than six or eight in a season."

PETREL, *Thalassidroma pelagica*.—Seen at North Unst in August, and at Dunnet Head on Oct. 10th, wind strong W., and haze, at 11 p.m.

TERN, *Sterna*.—Only one record. One struck at Pentland Skerries at 8.30 p.m. on Sept. 8th; wind strong S., and fog.

Terns always appear upon our Stirlingshire coast the end of August and beginning of September, remaining usually about a fortnight to three weeks.

The unusual scarcity of migrants is very generally reported upon at almost all the east coast stations. The remarks of the reporters, comparing the observations with former years, generally show this to be the case. Thus, commencing with our most northerly station. North Unst, I find it stated that, in most seasons, "Land birds, as Snow Buntings, Stonechats, and Starlings, strike the lantern." It is also stated that—"Until the last three years, large flocks of Iceland Gulls were seen during the month of April, going N.W. A few returned this way." They are seen, however, every year in some numbers in October. At Sumburgh Head, Mr. Anderson says, "In twenty-one years I have not seen so few birds strike the lantern"; and Mr. Tulloch, writing from North Ronaldshay, says, "We have had so much of N.W. winds here this fall, that I think the birds have been kept more

towards the south." Similar complaints of scarcity of migrants reach me from Auskerry, Hoy Sound (High), Holborn Head, Ness Head, and Isle of May; and an entire *absence* of all birds—"since the schedules came to hand"—is reported at several of these stations. The reason assigned by Mr. Tulloch is no doubt the correct one. From other stations local influences probably have more to do with the scarcity, as, for instance, the unfavourable positions of the lanterns. Thus, Cromarty is "situated on the point of the town of Cromarty, and almost among the houses." During seventeen years that the present principal light-keeper has been there, he does not think that more than a dozen birds have struck the glass. Altogether, few of our Scottish stations are situated in as favourable positions for observation as the lightships of the English coast. Perhaps the Bell Rock and Isle of May are the most favourable in the south, and the Pentland Skerries and some of the Orkney and Shetland stations in the north.

Besides the records under the various species which have been identified, there are many records of birds striking or passing, which were not identified. It is from a comparison of these and the other records that I arrive at the conclusion that the above stations are the most favoured. Thus, large numbers of birds struck the lantern of Bell Rock, and were lost between midnight and dawn on Oct. 7th, and again on the 14th, between 3 a.m. and dawn; on both occasions in foggy, hazy, or rainy weather, &c., as is almost invariably the case. I have not been able to trace from actual data, except in a few cases, the direction of the flight of the migrants, as such has not, as a rule, been noted by the reporters. It would be an advantage to have this done next season, under the column in the schedule for "General Observations." The species which have appeared in greatest numbers are Thrushes and Blackbirds, Larks, Wheatears, and Swallows; whilst Hooded Crows—noted as most numerous on the English east coast—are absent from all the Scottish returns, and Rooks occur only in one very far north instance, viz., at North Unst.

The line of flight of the *Anatidæ* is generally from N. to S., as explained further on by Mr. Cordeaux. I may mention here that this holds generally also on the west coast of Scotland, but that the lines of flight of the *Insessores* are influenced by local causes, and configuration and trend of the west coast. Thus the direction, according to previous experience, is often

from N.W. to S.E. When migrating over or near land the lines of flight follow the great valleys and then the coast-lines, often cutting off promontories, as at the Ross of Mull. These are probably birds having a starting-point more to the N.W.,—the Hebrides, Iceland, and North America,—and others which in their migrations from east to west overshoot the land and "hark back"; but we will speak more of this in our remarks under "West Coast." Land birds have passed at all hours of the day and night, flying generally low, and in almost all kinds of weather. The *Anatidæ*, as a rule, fly much higher, and consequently are only seen, or almost only, in clear weather.

Almost all records of birds caught or killed, or striking at the lanterns, are noted on dark or cloudy nights, with fog, haze or rain, or snow and sleet. The isolated stations, such as the Bell Rock, are most deadly; many are stunned and killed, and blown into the sea, at such localities. Birds on such nights often remain around the lights all night or rest on the window-sills of the tower and the balconies, or endeavour to obtain entrance to the tower. Whenever dawn appears they resume their flight towards the nearest land.

A great northward rush appears to have taken place at Bell Rock lighthouse between the 2nd and 11th of March, 1880, as noted in the duplicate schedule by Mr. James Jack, lately received by me. On the 2nd March great numbers of birds were seen flying about the lantern, apparently of various species, of which Blackbirds, Thrushes, and Mavises were recognised; wind S., light breeze, heavy sleet; sexes unidentified; a great number continued to strike hard on lantern all night from 10 p.m., and rebounding off, fell into the sea. On the 9th, great numbers, including Lapwings, Thrushes, Mavises, "Snowflecks," Rooks and "Sea-piets," from 5 a.m. till good daylight, when all disappeared in the fog; observer could not say in which direction they flew; wind S.E., light breeze, fog, and rain. This has been the largest migration seen this year; very large numbers of each species seen. On the 11th March great numbers of Lapwings, Curlews, Blackbirds, and "Sea-piets," from 12 midnight till first streak of day, when all disappeared; cannot say in which direction; wind N.E., light breeze, and haze. A few struck lantern-lights and flew off again; sexes not made out. A perceptible increase in our birds at inland localities took place about this time, and notes on migration taken at various inland localities corroborate the rush at coast stations. Curlews were seen passing inland on March 6th for first

time. Great Tits much increased in numbers, as also Chaffinches, Starlings, Thrushes, Blackbirds, &c.

EAST COAST OF ENGLAND.

Printed forms of enquiry and letters of instruction were sent to thirty-seven lighthouses and lightvessels on the east coast of England. Twenty-five stations have sent in returns, and twelve have failed to do so. The stations from which co-operation was asked are as follows, commencing with the most northerly. Those from which returns have been received are marked with a *.

Berwick.

*Longstone, on the most easterly of the Farn Islands; white light, revolving every 30 seconds.

*Farn Islands (two lights); white, revolving every 30 seconds.

*Coquet Island, coast of Northumberland; two white lights in same tower.

Souter Point.

*Teesmouth, No. 5 Buoy Lightvessel; one white light, fixed.

*Whitby, High Light; white, fixed.

Flamborough Head; two white, one red flash, revolving every half minute.

Spurn Point.

*Spurn, Lightvessel, six miles S.S.E. of Spurn Point; white, revolving every minute.

Bull, Lightvessel, mouth of Humber.

*Outer Dowsing, Lightvessel, 53 miles E.S.E. Spurn; red light, revolving every 90 seconds.

*Inner Dowsing, Lightvessel, 17 miles E. Sutton, on Lincolnshire coast; green, 20 seconds.

Dudgeon, Lightvessel, 35 m. E. Skegness, Lincolnshire coast; white, fixed.

*Leman and Ower, Lightvessel, 48 miles E.N.E. of Cromer; two white lights, one fixed, one revolving.

*Hunstanton, Lighthouse, white, fixed.

*Cromer, Lighthouse; white, revolving every minute.

Hasborough, Lighthouse; white, fixed.

Winterton, Lighthouse; white, fixed.

- *Newarp, Lightvessel, 12 miles N.E. Winterton; white, three quick flashes, and 36 seconds in a minute dark.
- *Cockle, Lightvessel, 3 miles E. Yarmouth; white, revolving each minute.
- *Lowestoft, Lighthouse; white, revolving 30 seconds, and red light, fixed.
- *Corton, Lightvessel, 4 miles E. Lowestoft; red, revolving 20 seconds.
- *Orfordness, Lighthouse; white, fixed.
- *Shipwash, Lightvessel, 8 miles S.S.E. of Orfordness; white, fixed.
- Languard Point, Lighthouse, entrance to Harwich River.
- Cork, Lightvessel, 5 miles off mouth of Harwich River.
- *Gallopier, Lightvessel, 40 miles S.S.E. of Orfordness; two white, horizontal, fixed.
- *Kentish Knock, Lightvessel, 33 miles N.E. of North Foreland; white, revolving each minute.
- *Nore, Lightvessel, mouth of Thames; white, revolving 30 seconds.
- *North Foreland, Lighthouse; white, fixed.
- *Goodwin, Lightvessel, Goodwin Sands; white, revolving, three quick flashes in succession, 36 seconds darkness.
- *Gull-stream, Lightvessel, Goodwin Sands; white, revolving 20 seconds.
- *East-side, Lightvessel, Goodwin Sands, 13 miles E. of Deal; green, revolving 15 seconds.
- *South-sand Head, Lightvessel, Goodwin Sands; white, fixed.
- South Foreland, Lighthouse; white, fixed.
- Dungeness Point; white, fixed.

Great credit is due to the observers of the various stations for the careful manner in which, as a rule, the returns have been made out. Taking them altogether the reports show truthful, accurate, and painstaking observation. Excluding the *Anatidæ*, Swans, Geese and Ducks, notes have been taken of about thirty-two different species.

SEPARATE REPORT OF EACH SPECIES OBSERVED.

Of *Raptores* two only are mentioned, the Sparrowhawk and the Short-eared Owl. Of the former one visited the Inner Dowsing on Oct. 1st, at 8 p.m. At the Gull-stream on Sept. 19th one alighted on deck and was caught. At the South-sand Head on Sept. 23rd a very large Sparrowhawk stopped to rest on the rail, going W. At the same station on Oct. 2nd another settled on the lantern, also leaving for the west. These may have been hawks in passage, or pirates beating in the track of small migrants. The Short-eared Owl occurs only at Teesmouth; on Sept. 29th one passed at daybreak, one on Oct. 1st at 7 a.m., and another on the 16th at 3 p.m.

I saw the first Short-eared Owl on the Lincolnshire coast on Oct. 30th. Twelve were seen at Spurn the same morning. It thus appears that the migration of this species has extended just over a month, from Sept. 29th to Oct. 30th.

FLYCATCHER, *Muscicapa grisola*.—At the Galloper on Sept. 13th, from 10 p.m. to 4 a.m., hazy, 200 to 300 birds, mostly Flycatchers; twenty killed against glass of lantern. At the Kentish Knock on Sept. 18th two dozen Flycatchers through night around lantern, rain, going S.W. at daylight; none killed. At the Nore, Aug. 22nd, at sunrise, mixed with Larks, and again on Sept. 7th, same hour, without Larks. At the Gull-stream, on Sept. 17th, at 3.20 a.m., E., thick rain, large quantities of Flycatchers; lantern surrounded by birds. On Oct. 11th-12th, 11 p.m. to 5 a.m., N.N.E., rain, large numbers of Larks, Starlings and Flycatchers in vicinity of light; many struck and went overboard. Time of migration extending over fifty-two days. ^[4]

[4] I find the following reference to this species in my note-book:—"Did not observe any Flycatchers, old or young, in this neighbourhood after the end of August: up to this time they were exceptionally numerous."—J. C.

SONG THRUSH, REDWING, FIELDFARE, BLACKBIRD, and RING OUZEL.—With the exception of the last, which is reported from Heligoland, the *Turdidæ* are noticed at fourteen stations extending over the whole coast-line. At the Longstone, on Sept. 11th, four Redwings killed at 3 a.m., gloomy and thick; on Oct. 30th and 31st Thrushes, Blackbirds and Redwings passed, wind E. and N. At the Farn Islands, Oct. 14th, Redwings, 9 a.m.; 15th, Thrushes and Blackbirds, N.N.E. half a gale; 22nd, several Blackbirds; 30th, same, and flocks of Redwings all day, E.; Nov. 20th, Fieldfares all day, E. strong; 22nd, same; Dec. 5th, 3 p.m., snow. Blackbirds and Thrushes. At Teesmouth on Dec. 1st, 10 a.m. to 1 p.m., snow. Redwings, Thrushes and Fieldfares,

going from N. to S.W.; Dec. 2nd, a few Redwings. At Teesmouth lighthouse on Oct. 1st, at daylight, forty to fifty Thrushes, and twenty to thirty at 9 a.m. on the 7th; on Dec. 4th, at 6 a.m., one Blackbird struck the glass and was killed. At Spurn on Dec. 2nd, 9 a.m., large numbers of Thrushes. At the Outer Dowsing, Sept. 24th, four Blackbirds; on the 29th, 10 a.m. to 1 p.m., six Thrushes. At the Leman and Ower on Dec. 16th, Thrushes in company with Larks and Starlings all through the night. At Cromer on Oct. 24th three Thrushes struck the lantern during the night; on the 30th four, and on Nov. 16th five others. At the Newarp, Nov. 28th, 9 a.m., six Blackbirds passed. At the Kentish Knock, Nov. 18th, one Thrush killed. At the Nore, Nov. 7th, four Blackbirds, going W.S.W.; Nov. 3rd, Fieldfares. At the North Foreland, Nov. 21st, 1 a.m., E.S.E., snow, one Blackbird killed. At the Gull-stream, Nov. 21st, cloudy and showers of snow, Starlings, Blackbirds, and Thrushes in the vicinity of light from 1 to 4 a.m.; six Starlings, four Thrushes, and one Blackbird killed. At the South-sand Head, Oct. 12th, dense fog, many Thrushes; several killed. General line of migration, with few exceptions, E. to W., carried on both by day and night; a considerable proportion, however, appear to arrive from daybreak to 10 a.m. Migration of Thrush extending over seventy-seven days, from Oct. 1st at Teesmouth to Dec. 16th at the Leman and Ower; of Redwing, sixty-five days, from Sept. 11th at the Longstone to Dec. 2nd at Teesmouth; of Fieldfare, twenty-eight days, from Nov. 3rd at the Nore to Dec. 1st at Teesmouth;^[5] of Blackbird, seventy-two days, from Sept. 24th at the Outer Dowsing to Dec. 5th at Farn Islands.^[6] On Oct. 29th, S.E. and E., *Tardus iliacus* crossed Heligoland in large numbers, "thousands and thousands" passing on overhead; the same day, *T. torquatus*, "a few"; *T. merula* and *T. musicus*, "limited numbers"; "*T. iliacus* descending from invisible heights down to about one hundred feet above island,—three hundred feet above the sea,—then all of them passing on; when atmosphere got clear again, in afternoon, passage ceased." Again on the night of the 30th, N.N.E., "enormous number of *Turdidæ* passing overhead."

^[5] Mr. W. Eagle Clarke reports that on the 24th of January last large numbers of Fieldfares were seen between Easington and Kilnsea, near Spurn Point, and as they were the first seen in the neighbourhood this season they were no doubt immigrants, more particularly as the locality in which they were seen is one where they are only observed during or immediately after arrival. Large flocks appeared also in North-East Lincolnshire during the last fortnight in January, so far as my own observations go, composed entirely of old birds. Mr. Gätke also

writes, "All January through, night from 10th to 11th, great many from the east; 21st to 24th, the same."—J. C.

[6] Numbers of fine old cock Blackbirds arrived at Spurn during the latter part of January, and early in February at the same time with the Fieldfares.—J. C.

REDBREAST, WHITETHROAT, WREN.—The same notes in the reports applying generally to the three species, it is not necessary to treat them separately. At Coquet Island, on Sept. 13th, 11 p.m., two Redbreasts struck glass, not killed. At Hunstanton, on Oct. 13th, 1 p.m., fog, a Wren and Redbreast seen. At Cromer, May 23rd, 1 a.m., a great many Wrens; two killed, also one Redbreast; May 27th, 11 p.m., Wrens and Whitethroats, five killed (these last notes are referable to the spring migration); Sept. 12th, Wrens and three Redbreasts strike glass. At Spurn this autumn the Common Wren did not appear before the middle of December; they usually arrive about the end of October.^[7] Redbreasts were very numerous at Spurn on Oct. 30th, wind blowing very fresh from N.E. None reported from Heligoland, but a great many *Accentor modularis* on Oct. 29th, wind S.E. and E.

[7] A friend, the owner of several steam-tugs, informs me that he has frequently known the Common Wren to come on board his vessels when far from land on the North Sea.—J. C.

GOLDEN-CRESTED WREN.—Most exceptionally scarce; only one mentioned, at the Cockle, on Oct. 16th, at 3 p.m., wind N.N.E., when one came on board and was caught. Three were seen in a garden near Spurn on Oct. 27th and 28th; none, so far as I am aware, on any part of the Lincolnshire and Durham coasts, neither are they mentioned in the Heligoland report.

TITMICE.—At the Gull-stream, Sept. 17th, 3.20 a.m., thick, rain, E., several Titmice seen round lantern amongst the Flycatchers. Mr. Gätke remarks, "There have been no *Parus major* all this autumn, and we have actually seen only one bird"; early in October, "daily some *Parus ater*, that come here rarely but when there is an appearance of easterly winds."

WHEATEAR.—At Hunstanton, May 24th, 11 p.m. to 2 a.m., drizzly rain, forty-one small birds killed, amongst them three Wheatears. At Cromer, Aug. 16th, 1 a.m., several Wheatears; four killed.

TITLARK, *Anthus sp.*?—At Spurn, May 25th, 1 a.m., rain, several Titlarks killed (referable to the spring migration), At the South-sand Head, Oct. 12th, 3 a.m., N.E., dense fog, Titlarks, amongst thousands of other birds, around and passing lantern.

SHORE LARK, *Otocorys alpestris*.—On Dec. 19th two were shot, from a small flock on the Humber side of Spurn Point; these were received in the flesh on the 22nd by Mr. W. Eagle Clarke, of Leeds.^[8] At Heligoland, on Oct. 1st, "some flights"; 5th, "some score"; 8th, "a great many"; 10th, "many flights"; on the 24th likewise, and again on the 29th, "many."

[8] Mr. Clarke subsequently informed me that on March 18th, this year, he saw a flock of about twenty Shore Larks at Spurn, and that during the past winter thirty-three have been shot there, in the proportion of two males to one female.—
J. C.

SKY LARK, *Alauda arvensis*.—None reported north of the Humber, and a negative return from Heligoland. South of the Humber, from every station making returns, and showing a gradual increase towards the more southern stations, where they crossed in immense numbers. Earliest date of passage, July 24th, at the Goodwin, 10 a.m., ninety to a hundred flying N.W. to the nearest land, and again at the same station on Aug. 4th. After this no further notice till Sept. 17th, when the regular and normal migration of this species appears to have set in and continued without intermission till Dec. 27th. Leaving out the two earliest dates as exceptional, we shall find that the immigration of the Larks to our east coast was continued over 102 days. During October, November and December (first fortnight) they crossed continually, passing from E. to W., also from points S. of E. to others N. of W., over the narrowest part of the North Sea, as is shown in the returns of the most southern lightships—Gallopier, Kentish Knock, Nore, North Foreland, Goodwin, Gull-stream, East Side, and South-sand Head; time, any hour of the day or night; direction of wind, variable. It would be quite impossible in a limited space to give anything like a detailed statement of the immigration of Larks, so a few extracts from my summary of the reports must suffice. There was a "great rush" of Larks and other species between Oct. 12th and 23rd. On the 16th October more birds appear to have crossed than on any day during the autumn. There was again a "final rush" of the laggards just preceding and during the outbreak of severe weather early in December. At the Newarp, Sept. 29th to Oct. 27th, on fourteen days, going W., no particular hour, some striking lantern at night. At the Cockle, Sept. 28th to Oct. 17th, as a rule, very early in the morning (4.30 to 9.20 a.m.), on foggy, thick nights stopping to hover round lantern, some strike and are killed; Oct. 9th to 16th, "great rush." At the Corton, Oct. 22nd and 23rd, flocks passing all day to N.W.; Nov. 11th, many hundreds during day; and

again on Dec. 25th, all day from N.E. to S.W., Ducks and Larks; Dec. 27th, hundreds of Larks during afternoon. At the Shipwash, from Oct. 9th to 16th; all day on 15th and 16th. At the Galloper,^[9] from Oct. 8th to 24th, Larks, Starlings and Chaffinches, generally during night; between these dates 135 of these species killed against lantern. At the Kentish Knock, Oct. 23rd, N.W., hazy, Larks and Chaffinches through night, "in clouds," sixty picked up on deck, and probably double this number falling overboard; Dec. 1st, from noon to 2.30 p.m., thick snow falling, Larks, Curlews, Starlings, Ducks, Chaffinches and Linnets, too numerous to mention; they came in clouds from N.E. to W.S.W. At the Nore, Aug. 4th to Dec. 6th and 7th, on forty-four days; on Dec. 6th and 7th, by hundreds all day, often mixed with Starlings, Linnets and Sparrows. At the North Foreland, on Sept. 17th, a few; Oct. 16th, midnight, misty, fourteen Larks and ten Starlings; 17th, between 3 and 4 a.m., ten Larks and nine Starlings struck. At the Goodwin, Sept. 17th to Dec. 7th, on nineteen days, going W. or N.W. or W.N.W.; Oct. 13th and 15th, in shoals throughout night; 13th to 22nd, "great rush." At the Gull-stream, Oct. 16th to 22nd, "great rush," flocks two hundred yards apart, passing westward. At the East-side, Oct. 9th to Dec. 12th, on seven days, E. to W.; on Sept. 23rd, going N. to S., with a N.E. wind blowing. At the South-sand Head, Oct. 12th, 3 a.m., dense fog, "great rush," thousands of Larks and other birds, many killed; Dec. 1st, 3 p.m., dense fog and snow, "final rush," Larks, Starlings, and fifty Curlews; twelve Starlings caught on board, but no Larks. Mr. J. H. Gurney, jun., writing from Lowestoft, says, "On Wednesday (Oct. 22nd), the arrival of Larks here was regular throughout the day." On Oct. 29th, at Spurn Point, Mr. W. Eagle Clarke, of Leeds, says, "Sky Larks arriving and passing south in thousands the whole day." Immense numbers also arrived on the Lincolnshire coast in the last fortnight of October. (At Heligoland, Oct. 29th, S.E., and E., "*Alauda arboreus*, many.")

[9] Birds passing this station were all passing to E. or E.S.E. or S.E. in October. On Nov. 24th great numbers of Larks and Starlings were observed passing to N.N.E. See general remarks at end of report.—J. C.

SNOW BUNTING, *Plectrophanes nivalis*.—None at stations south of the Humber. At the Longstone, Sept. 5th, 8.30 a.m., wind S., one was seen on the rocks; my correspondent remarks, "Earliest date I have ever seen a Snow Bunting on these islands (Farn Isles)." At the Farn Islands, Oct. 3rd, 11.30 a.m., three Snow Buntings flying N.W.; Oct. 25th, afternoon, in

flocks flying W. At Teesmouth, Nov. 7th to Dec. 20th, several flocks between these dates flying N.W. At Spurn, Snow Buntings were first seen on the 28th. October; and large flocks of from 400 to 500 in the Lincolnshire marshes during the first week in November. At Heligoland on Oct. 30th, N.N.E., "*Emberiza nivalis*, a great many," and on the 31st, S.E., calm, overcast, "Snow Buntings and Finches." Leaving out the Farn Island date as exceptional, ^[10] we find the regular migration of this species to have commenced on Oct. 25th and finished Dec. 20th—fifty-seven days.

^[10] In a note to Yarrell's 'British Birds' (vol. ii., p. 6, 4th ed.), Prof. Newton cites, on my authority, the occurrence of a Snow Bunting on the Lincolnshire coast on Sept. 16th, 1875, as perhaps the earliest date for England. The Farn Island bird must now take precedence by eleven days.—J. C.

CHAFFINCH, *Fringilla cælebs*.—Has occurred at several stations, and both north and south of the Humber. At the Longstone, Sept. 28th and Oct. 1st, 6 a.m. till noon, S. strong, rain and mist, flocks of Finches and other birds going W. At the Outer Dowsing, Oct. 18th, at noon, going E. to W.; Oct. 24th, 7 p.m., overcast, rain, thirty round lantern, two killed. At the Leman and Owers, Oct. 7th, N.N.E., gloomy, mixed flocks of Larks, Chaffinches and Starlings during day and night. At the Newarp, Sept. 26th to Oct. 23rd, on several days. At the Cockle, odd birds, on three days, strike lantern or come on board. At the Kentish Knock, Oct. 23rd and Dec. 1st (see [Sky Lark](#)). At the Goodwin, Sept. 17th, 9 to 11 p.m., overcast, hazy, flying round lantern. Migration extending from Sept. 17th to Dec. 1st—seventy-six days.

LINNET, *Linota cannabina*.—None registered north of the Shipwash, off the Essex coast, where from Oct. 9th to 16th flocks passed daily, all hours, going E. to W. At the Nore, Oct. 14th to Nov. 12th, invariably mixed with Larks. At the Gull-stream, Oct. 15th to 22nd, many flocks.

TWITE, *Linota flavirostris*.—One station only, the Kentish Knock, Oct. 24th, misty with rain, "Mountain Linnets," Larks and Starlings through the night; sixty picked up on deck, as many more went overboard. Nov. 8th, one caught on deck.

TREE SPARROW, *Passer montanus*.—Flocks of Sparrows^[11] were noticed at several stations south of the Humber. At the Outer Dowsing, on Sept. 29th, at noon, and Oct. 13th, 2 p.m., travelling from E. to W. At the Kentish Knock, Oct. 22nd to Nov. 8th, going from S.E. to N.W.; on Nov. 8th two

came on board. At the Nore, Oct. 27th and Nov. 15th, on latter day mixed with Larks. At the Goodwin, Oct. 6th and 22nd, passing to W.N.W. At the Gull-stream, Oct 16th, large numbers of Sparrows and Linnets going W.; also on the 24th, 3.30 p.m., flocks 200 yards apart. At the East-side, Nov. 8th, 9 a.m., a flock going E. to W. At the South-sand Head, Oct. 8th, "French Sparrows"; 16th, the same, some settled on board. Time of migration Sept. 29th to Nov. 15th—forty-eight days. During September and October Tree Sparrows usually arrive in North-East Lincolnshire in flocks containing many hundreds. This year has been quite an exceptional one, and only a few stragglers have been seen. I suspect the persistence of north-westerly winds in the autumn has driven these and the great body of our immigrants much further southward than is usual. The returns indeed show this to be the case.

[11] It is not improbable that in some of the returns the entries may refer to the Common Sparrow. We know, however, that *Passer montanus* is such a constant and regular migrant to our east coast in the autumn that in the majority of cases it will refer to this species.—J. C.

HOUSE SPARROW, *Passer domesticus*.—Only once mentioned, at the Corton, Sept. 28th, 7.50 a.m., large flock going west, males and females. As the female of the preceding is undistinguishable from the male, this entry undoubtedly refers to the Common Sparrow. We know that the House Sparrow is a migrant across Heligoland, [12] and my own observations on the Lincolnshire coast tend to the same conclusion.

[12] See a letter by Mr. Gätke in 'The Times,' March 31st, 1877, on the "Migration of Sparrows."

STARLING, *Sturnus vulgaris*.—Next to the Lark, the Starling occupies the most prominent position in the reports. It is noticed both north and south of the Humber at seventeen stations:—the Longstone, Coquet Island, Teesmouth, Outer Dowsing, Inner Dowsing, Leman and Ower, Cromer, Newarp, Cockle, Galloper, Kentish Knock, Nore, North Foreland, Goodwin, Gull-stream, East-side, and South-sand Head. Enormous numbers passed the southern stations in October, November and December. Sometimes in separate flocks, sometimes mixed with other birds. They crossed at all hours of the day and night, and in all winds and weather. Earliest recorded, Cromer, June 10th, 2 a.m., "quantity"; the Newarp, 7 a.m., on July 4th, going E. to W.; these were young birds. [13] Leaving out the earliest dates as referring to the migration of the young, we find that the

regular immigration on to our east coast commenced on Sept. 22nd, when four were killed against the lantern of the Longstone at 3 a.m., to Dec. 16th at the Leman and Ower—eighty-six days; migration E. to W. The Starling, like the Lark, appears to have a particular facility for immolating itself against the lanterns. At the Leman and Ower large numbers of Starlings, along with Larks and Chaffinches, were taken in October; at the Galloper (see [Sky Lark](#)); also at the North Foreland, and many other instances too numerous to mention. At Heligoland this year there have been no young Starlings during July; Mr. Gätke says he may perhaps have seen a hundred, and this is all.

[13] This bears out Mr. Gätke's observation in a letter dated Heligoland, 1879, when he says, "Hundreds of thousands of young Starlings from end of June to end of July. Hundreds of thousands of old birds during October, November, and later; young and old strictly divided in their migrations."

HOODED CROW, *Corvus cornix*; ROOK, *C. frugilegus*; DAW, *C. monedula*.—The migration of Rooks is not noted on the English coast north of the Humber; south of this they were noticed at nearly every station as either Rooks or "Black Crows," to distinguish them from the Hoodie or Grey Crow. Immense numbers of both Rooks and Grey Crows crossed on Oct. 16th from daylight to dark, travelling from E. to W. Noticeably at the Inner Dowsing, on Oct. 16th, large number of Crows, from N.E. to W.; at Hunstanton, on the 17th, Hooded Crows and Rooks all day; at the Newarp, on the 16th; at the Cockle, the same day "height of rush," Crows and Rooks, 9 a.m. to 6 p.m., going W., and again on the 22nd, 23rd and 24th, large numbers of Crows, Rooks, Starlings and Larks, from daylight to dark, passing W.; again on Nov. 8th, large flocks of Crows. At the Corton, on Oct. 16th and 22nd, continuous flocks of Crows, Daws and Larks all day, E. to W., also Nov. 8th and 11th and Dec. 27th. At the Shipwash, Oct. 15th and 16th, constant on 15th, till 12 a.m.; on 16th, 9 a.m. to 3.20 p.m., from S.S.E. to N.N.W. At the Goodwin, Oct. 15th to 21st, Crows "in shoals." At the South-sand Head, Oct. 16th, Rooks, 9.30 a.m., going W.N.W. and N.W.; and seven other stations. In the returns it is impossible in many cases, under the general entry of "Crows," to separate the Hooded Crow from the Rook. It appears, however, that immense numbers of the *Corvidæ* crossed the North Sea; Rooks at stations south of the Humber, and Hooded Crows at both northern and southern stations from Oct. 15th to Nov. 19th. Both Rooks and Hooded Crows appear to have started simultaneously on Oct. 15th, and

passed across the North Sea from E. to W. in almost continuous flocks on the 16th and 17th; and after these dates in more scattered detachments, less and less to Nov. 19th. And again during the latter, part of December; the last entry is on Dec. 27th at the Corton, Crows passing E. and W. in afternoon and Larks in hundreds. Daws are only mentioned at two stations; the Corton on Oct. 10th and 2:2nd, continuous flocks of Daws and Crows all day; also at the Shipwash on the 16th. At the Inner Dowsing on Oct. 22nd four Crows remained all night on globe above the lantern, and five took up their position in the same quarter on the night of Nov. 8th, leaving at daylight for the Lincolnshire coast. The wind on the North Sea on Oct. 16th was N. to N.E., force 4 to 7.

CUCKOO, *Cuculus canorus*.—On Sept. 17th, 10 p.m., at the North Foreland, wind E., overcast, misty a Cuckoo struck the south side of the lantern, but was not killed.

SWALLOW, *Hirundo rustica*.—The spring migration was noticed at two stations—the Cockle, June 8th, 9.20 a.m., five Swallows passed towards land; also on July 1st at 4.30 a.m., ten passing to W.S.W.; at the South-sand Head, July 6th, twenty at 3.30 a.m., flying towards N.N.W. The autumn migration was observed at several stations both north and south of the Humber. At Coquet Island, Aug. 15th, midnight, S.E., three Swallows struck glass of lantern, but were not killed; on the 20th great numbers were seen flying about, two striking glass after dark, but were not killed. At Whitby, on Oct. 4th, 9.30 a.m., flock going south. At Hunstanton, on Sept. 10th, 2 p.m., a flock alighted on lantern and gallery; and on Oct. 12th, 3 p.m., fog, twelve came up to light and dispersed in various directions. At the North Foreland, on Sept. 9th, 10th and 11th, about forty passed each day; and on the 21st, about one hundred flew round the lantern top for an hour. At the South-sand Head on Sept. 28th, twelve. Swallows passed. Migration extending from Aug. 15th to Oct. 12th—fifty-nine days.

SWIFT, *Cypselus apus*.—The autumn migration of Swifts, so remarkable in 1879 for the great numbers seen at various places on the north-east coast (see Zool. Jan. 1880, p. 8), was observed at only two stations. At Teesmouth, great numbers flying about during the last three days of August. At the Nore, on Aug. 1st and 2nd, between 6 and 7 a.m., and again on the 20th; last on Oct. 10th, 9 a.m., Swifts mixed with Larks. At Heligoland,

Martins and Swifts, "rather great numbers during August; particularly noted on the 10th, wind S.E., Swifts a great many; on the 20th, the same."

CHARADRIIDÆ.—The burst of severe weather during the early days in December brought an immense influx of Plovers to our east coast from the north; large flocks are noticed at this time at the more northern stations, more particularly at Coquet Island on Nov. 26th and Dec. 6th, northerly and westerly winds, force from 6 to 8. There is only one notice south of the Humber, and this early in the season. At the Shipwash, Oct. 12th, flock of thirty Lapwings, going westward.

WOODCOCK, *Scolopax rusticola*.—Was observed at the Farn Islands, Oct. 22nd, 30th (great flight), Nov. 1st, Dec. 2nd, N.E. to N.N.E. At Coquet Island, Sept. 29th, S.W., strong, one; and Oct. 24th, also one; about noon on both days. At Teesmouth, Nov. 2nd, three at daylight. At the Outer Dowsing, a station more than fifty miles from the nearest land, on Dec. 12th, two flying W. At Heligoland, Oct. 30th, N.N.W., early at dawn, great many Woodcocks; about 100 killed. Great numbers were shot on the Yorkshire and Lincolnshire coast on the morning of Oct. 31st, sixty at Spurn Point alone; wind very fresh from N.E. on the previous night with drizzly rain. The migration of Woodcocks extended from Sept. 29th at Coquet Island to Dec. 12th at the Outer Dowsing—seventy-five days. The "great flight," at Heligoland on the morning of Oct. 30th; east coast of England, on the night of the 30th.

SNIBE, *Gallinago media*.—At the Longstone, Sept. 17th, 2.30 a.m., misty, one struck the lantern and was killed. At the Farn Islands, on Dec. 2nd, 10th and 19th. At the North Foreland, on Nov. 14th, 1.30 a.m., N., moderate gale, and again on the 21st, at 5 a.m., snow. Snipe were killed. Very large numbers arrived in Holderness and North Lincolnshire, from the middle to the end of November, just preceding the outbreak of severe weather in December. They left the district again, almost to a bird, before the middle of the month.

CURLEW, *Numenius arquatus*.—Curlews were in large flocks at Teesmouth on Sept. 27th. At the Inner Dowsing, Sept. 1st, three going from N.E. to S.W. At Hunstanton, on Oct. 13th, 4 a.m., fog, one struck glass of lantern and was taken. At the South-sand Head, Aug. 13th to 29th, large flocks, 100 to 200 in a flock, going S. and S.W. The Curlew migrates early in the

autumn, Aug. 13th to Sept. 29th. Most of these entries will probably refer to young birds, the old Curlews coming later in October and November. At Heligoland, Oct. 30th, many passed over.

ANATIDÆ.—Between Dec. 4th and 20th Swans passed the Teesmouth from N. to S. in some numbers; on the 4th, six; the 8th, five (four old and one young); 14th, nine (eight old and one young); 15th, forty-three; 16th, three; 18th, two; the 20th, ten at 10 a.m., and ninety-five at 11 a.m. On the 18th a flock of thirty was seen near Spurn, one old white bird acting as pilot, the remainder all young, in the brown plumage. Two Bewick's Swans were shot on the Humber on the 8th. Four passed the Kentish Knock, on Dec. 4th, 2.15 p.m., N.E. to S.W. Two Swans, which may have been "escaped birds" from some private waters, passed the Teesmouth on Sept. 30th, 10 a.m., flying S.E. Geese were observed at several stations both north and south of the Humber, going south from Aug. 18th to Dec. 13th, the main body passing south during the last half of November and in December. Brent or "Black Geese" are noted at the Corton, going south, July 18th, twelve; July 30th, two flocks of from thirty to forty; on Sept. 17th, thirty. These are very early dates for Brent Geese, ^[14] Mr. Cotton, the observer, might have possibly been led into an error by passing flocks of black Scoters. As, however, "black ducks" are frequently mentioned in his return, this could hardly be the case, and he appears very well able to discriminate between the two. At the Corton, Dec. 4th, hundreds of black geese from N.E. to S.W. from sunrise to noon—*i.e.*, towards the mouth of the Thames or Essex coast. As might have been expected, large flocks of Ducks are constantly noted, so frequently indeed, as to defy repetition; this is more particularly the case in the returns from the lightships. They are noted sometimes flying S., from the land or to the land, and in fact in every direction. Many of these would be local flocks going to and from their feeding grounds. Immense numbers, however, coming from the north, passed along the coast in November and December. At Coquet Island, on Dec. 14th, 3 a.m., N.W., blowing a gale, two Eider Ducks (spelt phonetically, "Ida" in the report) struck lantern and one was killed.

^[14] Major C. Russell, writing from Essex, says, "The earliest occurrence of Brent Geese on our coast I ever remember hearing of was Sept. 29th, when many years ago one of our coast gunners, still living, killed ten at a shot."

RED-THROATED DIVER, *Colymbus septentrionalis*.—Red-throated Divers and Mergansers are mentioned in the report from Teesmouth during November. With regard to the first of these, Mr. Gätke sends the following curious note, dated Dec. 22nd, 1879:—"Colymbus septentrionalis, almost by the million. During the last forty years there have never, during a single autumn, been a fiftieth part of what we see here now every day—all passing along, principally during the forenoon, east of the island in an E. by N. direction, which I think they continue till coming to the Holstein coast, then strike off in a northerly course up to the extreme north of Jutland, and from thence cross over to the Dutch coast, perhaps next morning to renew the trip. There are constantly so many that one scarcely can believe them always to be fresh birds."

PROCELLARIDÆ.—At the Longstone, on Sept. 18th, at 3 a.m., two Petrels were caught against the lantern-windows. At Coquet Island, on Nov. 4th, 3 a.m., one killed against the lantern. At the Goodwin, Oct. 11th, five were seen with many Larks and Starlings round the lantern at night. Greater numbers of *Procellaria glacialis* have been seen off Heligoland during the autumn than has been the case for the last twenty years.

Independent of the notes of each species, as already given, numerous flocks of small birds were seen passing the stations, but too far distant to determine the species. This was specially the case at the Spurn, on Dec. 3rd and 8th; at the Cockle, on Oct. 9th; and at the South-sand Head, on Sept. 20th. At the Kentish Knock, on Nov. 20th, half-a-dozen small birds came on board at midnight, which, from the written description in the margin, were probably Greenfinches. It is, however, impossible to give even a guess at another species which visited another lightvessel on Sept. 20th, "Fifty very small birds—in fact, the smallest of British birds—like a Sky Lark, but a deal smaller." From the date, they may have been either Titlarks or Flycatchers.

No rare migrants have been noticed, but this was scarcely to be expected. In this respect the budget of notes supplied by that veteran ornithologist Herr Gätke bears a striking contrast to our east coast reports. On that small island, so favourably situated for observation, Mr. Gätke has trained up

quite a host of practical observers, and any rare visitant will have to be very sharp if it succeeds in escaping detection. Quoting from Mr. Gätke's letter dated Dec. 22nd, we have, on Sept. 17th, *Sylvia fuscata*, though not got; *Anthus Richardi* on the 18th; *A. cervinus* on the 20th; *Picus leuconotus* on the 21st; then some days S.W. with rain and no birds; 26th, calm and clear, *Emberiza pusilla*; 27th, E.N.E., *Anthus Richardi*, *Emberiza pusilla* (two shot), *E. aureola* (seen); 28th, *E. pusilla* (two seen), *E. rustica* (one shot); 29th, *Sylvia superciliosus* (one); 30th, *Emberiza pusilla* (one); Oct. 1st, *E. pusilla* (one); Nov. 2nd, 3rd and 4th, westerly winds and nothing, but daily some *Parus ater*; on the 5th, again, *Anthus cervinus* (seen and heard); northerly winds, sometimes N.E. on the 8th, a Leaf Warbler seen at close quarters, but not procured by Aeuckens, the Heligoland birdstuffer, which Mr. Gätke feels tolerably certain was a very rare visitant; the same day there were two or three *E. rustica*, and one each of *A. pusilla* and *A. cervinus*. The rare Leaf Warbler was again seen on the 9th, but could not be obtained; an *Emberiza rustica* was also seen. On the 10th, *E. pusilla* shot and *E. rustica* seen. On the 12th, *Muscicapa parva* and *Motacilla lugubris*, which latter comes there rarely in autumn. On the 14th, *Sylvia superciliosus* was shot by Mr. Gätke's eldest son, and two more were seen by Aeuckens in a garden where shooting was not permitted. On the 18th, one very fine *E. pusilla* shot. On the 20th *Lestris affinis* was seen close by, but not shot, as it would have fallen in the sea and drifted away. This is the third, if not the fourth, known occurrence of the Siberian Herring Gull at Heligoland. On the 24th a large Redpoll was shot, which Mr. Gätke thinks can only be the Greenland bird, *Linota Hornemanni*.

A summary of the various returns show that four species, Larks, Starlings, Rooks and Hooded Crows, in the order given respectively, far outnumber any other, and of these four the Lark far exceeds the rest in migratory numbers. As a rule, the lines of migration followed have been from E. to W., or S.E. to N.W. direct on to the English coast. To the middle of October we find birds rarely coming from points north of east; after this date they appear to come occasionally from directions north of east, between east and due north. The line of migration of the *Anatidæ* has been from north to south, crossing the line of the *Insessores* at right angles.

There is one remarkable exception to the general rule of an east to west route in the case of the lightvessel on the Galloper Bank, forty miles S.S.E.

of Orfordness. All entries in this return show birds passing to the E. or S.S.E. or S.E. during October. We can only account for this anomalous line by supposing the birds which cross the Galloper have previously struck the English coast at some higher latitude, and after skirting the coast line shot off again somewhere near Orfordness and crossed the North Sea to the Belgian and French coasts between Nieuport and Dunkerque. On Nov. 24th a great quantity of Larks and Starlings passed the Galloper towards the N.N.E, a line which if persevered in would eventually bring them to the Dutch coast.

At the East-side from the 2nd to the 5th of November flocks of "Crows" were observed passing from the S. to the N., or from the French to the Suffolk coast. The great body of the immigrants during 1879 have crossed at the more southern stations, the four lightships on the Goodwin Sands, also the Nore, the Shipwash and Kentish Knock showing a constant and continuous stream on to the south-east coast. These birds would all cross at the narrowest part of the North Sea directly from E. to W. to the English coast. The long prevalence of north-westerly winds in October and November may to a great extent have deflected the lines of migration much farther to the south than is generally the case.

Migrants have passed the stations at all hours of the day and night, flying at no great altitude and in almost all winds and weather. When the nights are dark and cloudy, no stars appearing, in rain, fogs and snowstorms, flocks of birds during the night migrations will crowd round the lanterns of the lightships; many strike the glass and are killed, falling on deck or pitching overboard. On these nights birds will often remain for hours in the vicinity of a light, circling round and round, evidently having lost their way; at the first break in the clouds, the stars becoming visible, or the first streak of early dawn, they will resume their flight to the nearest land.

The "great rush" of immigrants was from the 12th to the 23rd of October. The "final rush" took place just preceding and during the first burst of winter, early in December.

The occurrences when birds travel dead to windward are very rare indeed, and then only with light winds. In the great majority of cases birds migrate flying within two to four points of the wind, sometimes with a "beam" wind, or even a point or two "abaft of beam." If the wind changes during

the actual passage, birds will change the direction of their flight to suit the wind. Thus at the Shipwash, on Oct. 16th, from 9.50 a.m. to 1.20 p.m. the wind was N.N.W., flocks constantly passing from S.E. to N.W. from 1.20 to 3.20 p.m. the wind blew from N.N.E., and the birds passed S.S.E. to N.N.W. If subsequent observations should prove the correctness of this rule, it will go far to account for all the irregularities of migration—the "why" birds are seen in great numbers in one year in any locality, and perhaps absent altogether in following seasons.

Subsequent to the writing of this report I have received several communications showing a large and very remarkable influx of immigrants on to the east coast of England in January and the first fortnight in February, 1880. On Jan. 24th great numbers of Fieldfares were seen near Spurn Point, between Easington and Kilnsea, and since this date they are reported as very numerous in the locality; with the Fieldfares came many fine old cock Blackbirds. In North-East Lincolnshire, also, during the last fortnight in January I saw many very large flocks of Fieldfares, apparently entirely composed of old birds; before this time scarcely any had been seen. Mr. Gätke writes from Heligoland:—"T. *pilaris* all January, through night from 10th to 11th, great many; also Oystercatchers and other waders coming from the E; also from 21st to 24th, the same." These dates agree very well with the time of their arrival on our east coast. Short-eared Owls arrived in some numbers at Spurn in the last week in January. A very large flock of Snow Buntings was seen on the 8th February. On the Lincolnshire side I noticed an immense flock of the same species, fresh arrivals, on the 5th; and on the 11th thousands, also hundreds of Larks. This final and third "rush" of immigrants so late in the winter is curious, and may perhaps be explained by the comparatively milder winter over Northern Europe, compared with what we have experienced in the central and southern zones of Europe. We have indeed a recent precedent for this in the winter of 1877-78, when, after a long continuance of mild weather in Scandinavia, a sharp and sudden outburst of real winter drove the Fieldfares, as Mr. Gätke says, over Heligoland, on the night from the 27th to 28th January, in "countless flights."

WEST COAST OF SCOTLAND.

Printed forms of enquiry and letters of instruction were forwarded to thirty-eight lighthouses on the west coast of Scotland and the Isles. Twenty-four stations have sent in returns, and fourteen have either sent none, or have returned blank forms, owing to the scarcity of birds. The stations from which co-operation was asked are as follows, commencing with the most northerly and those from which returns have been received are marked with a *:—

- *Cape Wrath, Sutherland; white and red alt., rev. every minute. J. M'Gill.
- *Rhu Stoir, W. Cromarty; white, interval 1 minute. William Wither
- *Butt of Lewis, Lewis; white, fixed. George Edgar
Stornoway, Lewis; 2 lights, white, rev. $\frac{1}{2}$ m., & white, fixed. R. Murray
- *Island Glass, Harris; white, fixed. William Lines
Ushenish, North Uist; red, fixed
- *Monach Island; same tower, upp. white fl., low. red, fixed. J. Youngclause
Barra Head, Barra; white, interval $2\frac{1}{2}$ minutes
Rona, Skye; white fl., every 12 seconds. David Dunnett
Kyleakin, Rosshire; white and red, fixed. David M'Culloch
- *Isle Ornsay, Skye; white, fixed. David Ross
- *Ardnamurchan Point, Argyleshire; white, fixed. David Laidlaw
- *{Hynish, Skerryvore Signal Tower, Tyree; no lights.† William Crow
{Skerryvore, off Tyree; white, revolving every minute. "
- *Dhuheartach, S.W. of Ross of Mull; white, with red sector, fixed.
J. Ewing
- *Sound of Mull, Tobermory; red, green, & white sectors, fixed.
W. M'Lellan
Corran Ferry, Loch Eil; red and white sectors, fixed. Thomas Tulloch
Lismore Island, Oban; white, fixed
- *Fladda, Easdale; white, with a red sector, fixed. James Langton

- *Rhuvaal, Islay; white, with red sector, fixed. David Spink
- *M'Arthur's Head, Islay; white and red sectors, fixed. Thos. Sutherland
- *Skervuile, Jura; white, revolving every ½ minute. John Ewing
- *Rhinns of Islay, Islay; white fl. every 5 seconds. Andrew Lyall
- Lochindaul, Islay; white and red sectors, fixed. William Mail
- Mull of Kintyre; white, fixed. James Thompson
- Sanda, Kintyre Sound; red, fixed
- *Devaar, Kintyre; white, revolving every ½ minute. Charles Black
- *Pladda, Arran; 2 lights 8 yards apart, white, fixed. Thomas Grierson
- *Lamlash, Arran; green, fixed. David Waters
- Turnberry, Ayrshire; white fl. every 12 seconds. Charles Black.
- *Corsewall, Wigtown; white and red alternate every minute. R. Laidlaw
- *Loch Ryan, Wigtown; white, fixed. Ralph Ewing
- *Portpatrick, Wigtown; white, fixed. James Beggs
- *Mull of Galloway, Wigtown; white, intervals. M. Morrison
- *Little Ross, Kirkcudbright; white fl., every 5 seconds. W. A. Mackay
- Point of Ayr, Isle of Man; white and red alternate every min. J. Blythe
- Douglas Head, Isle of Man; white, fixed. Alexander M'Donald
- Chickens Rock, Isle of Man; white, revolving every ½ minute
- † This station included by mistake, but retained for convenience, bracketed with the next.

Thanks are due to the various observers for the careful way in which the schedules have been filled in. Excluding the Natatores, notes have been taken on about thirty species.

SEPARATE REPORT OF EACH SPECIES OBSERVED.

HAWK.—At Dhuheartach on Oct. 5th, two struck lantern at 8 a.m., light S.E. wind, and haze; and another on the 8th, at 7 a.m., calm, with fog. At Mull of Galloway, on July 18th, a Sparrowhawk was killed at 11 p.m., S.E., light

and calm, thick haze; and one on Aug. 15th, at 1 a.m., wind variable, light air, fog and haze.

OWL.—I have only two records—one seen at Dhuheartach on May 30th, at 4 p.m., wind N.W., fog; and the other at the same place on Oct. 20th, at 3 p.m., wind E.S.E., fresh, with haze.

SWALLOW, *Hirundo rustica*, L.—The most northerly station noted this year on our west coast is Rhu Stoir, but two records here evidently refer to the spring migration, being June 6th and 10th, when two were seen on each day. The most northerly station noted during the autumn migration is Ardnamurchan; thence southwards at Skerryvore, Dhuheartach, Rhuvaal, Skervuile, Rhinns of Islay, and Loch Ryan. The earliest date on which they were recorded is 29th July, when one was seen at Dhuheartach at 6 p.m.; wind N., fresh, with clear weather. The latest record is Sept. 25th, when forty or fifty were seen at Rhuvaal, at 4 p.m.; wind S.W., fresh, with rain. The length of time thus occupied by the migration along the west coast of Scotland was, approximately, fifty-eight days. The greatest rush took place between the 20th Aug. and 25th Sept.; and, as far as shown, at Ardnamurchan and Rhuvaal. At Ardnamurchan, twenty rested all night on Aug. 20th, and left in the morning. At Rhuvaal forty to fifty were seen, as related above. Swallows appeared to migrate principally when the wind is southerly (S.W. to S.E.), in light, variable, or fresh head- or side-winds, and hazy weather. Exceptions are found, however, at Dhuheartach on July 26th, when the wind was N., fresh, and weather clear; and at Skerryvore on Aug. 17th, when one appeared at the lantern about 10 p.m. (two hours after lighting up) in a N.W. light breeze, with showers. Swallows migrate mostly by day, resting all night. My returns show that they passed at all hours, but mostly before dark.

MARTIN, *Hirundo urbica*.—Martins are noticed at two stations only, viz., Butt of Lewis and Corsewall. The migration lasted between July 29th and Aug. 16th, or during eighteen days (but data are defective). They travelled during the day, resting around the lighthouses at night, with S. to S.E. and S.W. winds; the only one killed was when wind was N.N.E., at Corsewall.

WREN.—The records are few of this species: usually noticed on migration in large numbers; they appeared at Monach Island and Dhuheartach. The earliest date is Aug. 17th, when "some" struck at 10 p.m., and between that

and 3 a.m.; wind W., and haze. The latest date is Dec. 9th, when one was seen at 3 p.m., with S.W. breeze, and showers. In fourteen years Mr. Edgar never saw a Wren at the Butt of Lewis lighthouse.

GOLDCREST, *Regulus cristatus*.—At Mull of Galloway on July 18th, alighting all night and resting. Mr. Edgar's letter of 20th Sept. supplies the information that he has never seen Goldcrests at the Butt of Lewis—a negative fact of value in tracing their exact lines of flight. Perhaps in our next reports observers will state in their remarks if any of the species noticed in this one have hitherto been rare, or entirely absent from their stations in past years.

ROBIN.—Appeared at Kyleakin, and thence southwards at Skerryvore, Rhuvaal, Skervuile, Rhinns of Islay, and Corsewall. The largest number recorded at Corsewall, where fifteen struck on Sept. 15th, at 11 p.m.; wind S.W., light breeze, and haze. The earliest date is Sept. 10th—one at Skervuile, at midnight; wind S.W., light, and cloudy. The latest is at Rhuvaal, on Dec. 4th at 9 a.m., when one was seen; wind variable, light breeze, clear. Thus, time occupied is between Sept. 10th and Dec. 4th. A rush took place in September between 10th and 21st, but no large numbers are recorded; in October another rush between 7th and 18th; on the former date numbers migrated with Larks and other species not known, and many were killed and fell into the sea at Skervuile. At Skerryvore one seen at 7 p.m.; wind S., fresh, with fog and rain. In September I have records on six days at four stations; in October two at two stations, and in December one at one station—Rhuvaal; they passed or struck mostly at night or morning, but a few through the day; winds mostly southerly, and hazy or foggy; a few in cloudy and clear.

HEDGESPARROW.—At M'Arthur's Head, on Sept. 28th, one struck at 1 a.m.; wind N.W., strong, with haze and rain.

WHEATEAR.—Wheatears or "Stonechecks" are recorded from Skerryvore as the most northerly station this year, and thence southward at Rhuvaal, Skervuile, Turnberry, Corsewall, and Mull of Galloway. The earliest appearance noted is on Aug. 15th, and the latest on Sept. 28th; thus the migration extended over forty-four days. The greatest rush was between Aug. 17th and 22nd. The migration took place entirely at night, and principally with southerly winds—S., S.W., and to W.; fewer in northerly or

easterly winds. The favoured stations were Skerryvore, Rhuvaal, and Skervuile.

TITMICE.—The negative information that Mr. Edgar never saw any Titmice at the Butt of Lewis lighthouse is worthy of record. I have notes of their occurrence at M'Arthur's Head and Lamlash; in each case a single bird struck the lantern, Nov. 16th and Aug. 17th, at 6 p.m. and 10 a.m. respectively; both during west wind, and light air.

FIELDFARE.—Occurred at Monach Island and Douglas Head, and is reported as usually seen at Butt of Lewis, though not this year. At Monach Island a number struck, and fifteen were killed, between 8 p.m. and 4 a.m., on Oct. 21st and 22nd; wind S.W., fresh, and drizzling rain. At Douglas Head one was seen on Oct. 16th, at 10 a.m.; wind N., moderate breeze, and cloudy.

THRUSH.—The most northern station is, for 1879, Isle Ornsay. Mr. Edgar; however, adds the general information that in most years the Butt of Lewis yields large numbers at the lantern; thence southward at Skerryvore, Dhuheartach, Rhinns of Islay, Lochindaul, Lamlash, Corsewall, and Point of Ayre. The earliest date is Sept. 19th, when a mixed flock of Thrushes and Blackbirds passed in variable winds, and clear, at Lamlash. The latest date is Dec. 3rd, when "Thrushes" passed Lochindaul in the "daytime." But this may be due to local influences and hard frost then registered. The next latest date is Nov. 24th, when one was seen in the daytime, 2 p.m.; wind N.E., fresh, clear; this would give sixty-six days for the passage, or, taking the later date of Dec. 3rd, seventy-six days. The rushes were on Sep. 19th (*ut sup.*); a "great rush" Oct. 21st to 28th. On Oct. 21st at Point of Ayr, two hundred "along with Blackbirds" seen at 1.80 p.m.; wind N. and N.W., strong, clear, with occasional showers. Also at the Rhinns of Islay, five seen, and so through records of Oct. 22nd, 18th to 28th, when three hundred "dark grey birds larger than a common Thrush!!?" (probably Missel Thrush, *Turdus viscivorus*) were seen at Point of Ayr; wind S.S.E., light, and hazy; and at Skerryvore about forty (of which ten killed) at 7 p.m.; wind S.E., light breeze, and haze. Thrushes travel with winds from S.E. to S.W. by preference, but also with winds from W. to N.E., through N.; observed most, as with most other species at lanterns, in hazy and rainy weather, but the great rush took place in "clear, with occasional showers," and "hazy and fine." Returns show that out of sixteen dates reported upon, only three show

a diurnal migration, but one of these three is the rush of three hundred "dark grey birds larger than a common Thrush"^[15] on Oct. 28th, at Point of Ayr.

^[15] We shall refer to these under "Birds unknown," but we have thought it advisable to enter them here also; they can easily be erased in future comparisons if better data come to hand.

BLACKBIRD.—The northernmost locality is Rhuvaal, where one struck on Dec. 1st, at 2 p.m., wind N.E., fresh; thence southwards at M'Arthur's Head, Skervuile, Pladda, Lamlash, Corsewall, Mull of Galloway and Point of Ayr. The earliest date is Aug. 26th, at M'Arthur's Head. The latest date is Dec. 1st, at Rhuvaal. The time thus occupied by the migration was ninety-seven days. The rushes took place between 13th and 21st Oct., and the most favoured locality recorded is Point of Ayr, where on Oct. 21st two hundred "mixed Thrushes and Blackbirds" were seen at 1.30 p.m.; wind N.N.W., strong, and clear, with occasional showers. They were visible at the various stations in all winds, apparently from W. to E. and from S. to N.N.W. They appeared at all hours of the day and night, but most of the records apply to night migration.

STARLING.—The records of the Starling are very general; the most northern station given is Butt of Lewis, where the remark is made that they generally strike. This year, however, the most northern is Ardnamurchan Point, thence southward at Dhuheartach, Sound of Mull, M'Arthur's Head, Skervuile, Rhinns of Islay, Corsewall, Portpatrick, Mull of Galloway, and Point of Ayr. The earliest date on record is Aug. 11th, at M'Arthur's Head and Loch Ryan; at the former, one struck at 10 p.m., in S. wind, fresh, with haze; at the latter, two struck at 11.30 p.m.; wind E., light, with fog. The latest date is Dec. 18th, when one was caught at Skervuile at 11 p.m.; wind S.W., fresh, with haze. Thus the time occupied in passage was one hundred and thirty days, but the last date may have been due to local influence and keen frost at that time registered. The next latest date is Nov. 23rd, which would make the time spent in passage one hundred and four days. Rushes took place more frequently than with most species noted; thus, Sept. 19th, Oct. 12th to 25th, and Nov. 11th and 17th, may all be characterised as dates of rushes. The most favoured localities were Corsewall and Portpatrick, for frequency of records; and Ardnamurchan Point, Dhuheartach, Portpatrick, and Point of Ayr, for numbers of individuals. Starlings do not seem to have any choice of wind, as far as returns show. We find them migrating with N., W., and S.E.

winds in almost equal proportions, with a preference—but very slight—to W. winds. They also pass or strike in E., S., and N.W. winds, but in fewer numbers. We do not think any deductions can be drawn from these data. But Starlings breed at many of the lighthouses or in their vicinity, so it is not always easy to determine whether, in every instance, they are on migration or not (as remarked by more than one reporter, *e.g.*, Mr. Edgar, at the Butt of Lewis, &c.). Out of seventeen dates reported on, only three show migration to have taken place during the day, *viz.*, at Portpatrick (in two instances), and Ardnamurchan Point (3 p.m. in a gale).

FINCHES.—At Dhuheartach on Aug. 24th, one hundred were seen and twenty killed between p.m. and 2 a.m., wind N.N.W., with haze; on Oct. 4th, six seen in afternoon, wind light and variable, with clear weather; again, on Oct. 8th, two were killed at 7 a.m., calm, with fog. The above are all the returns of "Finches."

LINNET.—Only two records. At Rhuvaal on Sept. 9th, thirty to forty old and young, were seen around the lantern at 10 a.m.; wind E. to N., breeze to stormy, with haze; and at Lamlash on same date and at same hour, fifty were seen; wind fresh breeze from W., and clear.

YELLOWHAMMER.—At Lamlash on Aug. 17th, one struck at 10 a.m., in light W. wind, and fog.

SNOW BUNTING.—At Monach Island, farthest north recorded, and at Dhuheartach and Corsewall. Earliest appearance Sept. 19th, when a flock was seen at Monach Island at noon, with S.W. light wind, and clear weather. Latest date Dec. 10th, when one was caught at Corsewall at 10 p.m.; wind variable, light, and haze; thus, time occupied was eighty-two days. A. rush took place probably about Sept. 19th, when a flock was seen at Monach Island,—but our data are scanty,—and another about Nov. 10th, when a flock passed Corsewall. Other dates of record are Oct. 10th and 18th; they passed in N.W. and N. winds, in clear or cloudy weather; also in September in S.W., with clear weather, and variable wind, with clear or haze. All the records are during the night, except one at 12 noon, at Monach Island on Sept. 19th; wind S.W., clear.

LARK.—The most northern station noted this year was Dhuheartach, and thence southward they are noticed at M'Arthur's Head, Skervuile, Rhinns of Islay, Lamlash and Corsewall. The earliest date is Aug. 17th, when twenty

struck at Dhuheartach between 10 p.m. and 3 a.m.; wind W., with haze. The latest date is Dec. 6th, when one struck at 9 p.m.; wind S.E., fresh, and haze, at Corsewall; the time thus occupied was one hundred and twelve days (but the last was no doubt due to the severe frost inland at that time). The previous date is Nov. 18th, which would make the time only seventy-nine days. The greatest rush appears to have been in October, about the 12th to 24th, but the numbers observed are so small that it is difficult to judge. The locality most favoured was Dhuheartach, whence I have returns on eight different dates, between Aug. 17th and Nov. 24th. An earlier rush took place about Aug. 17th to 21st, at Dhuheartach and Skervuile. A later rush about Nov. 15th to 18th at Dhuheartach, Skervuile, and Rhinns of Islay, but of small dimensions. Larks appear to migrate principally when the wind is W. to S.W., or late in the season, when still more southerly, say S. Exceptions occur at Dhuheartach on Nov. 24, when wind was N.E., fresh, and weather clear; at Rhinns of Islay on Nov. 7th, wind N.W., light breeze; at Skervuile on Oct. 10th, when wind was N.; they migrate in calm, variable, light, or fresh winds, and are noticed at lanterns principally in haze and fog. They migrate mostly at night, at all hours, but the majority pass between 7.30 p.m. and 4 a.m. Those passing during the day were travelling when the wind was N. or N.E.; those passing at night when the wind was S. to W. So the returns show; but whether this indicates any *law of migration* or not remains to be proved by further observation.

LAPWING.—The most northern station which they were observed to pass at this season was Rhu Stoir; thence southward, they are reported at Isle Ornsay, Lochindaul, Devaar, Pladda, and Portpatrick. The earliest date given is Aug. 20th, when "flocks" were seen during the day in N. wind and clear weather; the latest is Dec. 11th, when one was killed at Pladda, at 2 a.m., wind S.E., light breeze, with haze. The next latest date given is Nov. 14th, when twenty-two were seen at 8.30 a.m. at Portpatrick, wind S.E., and haze. The time occupied, therefore, this year is from eighty-six to one hundred and thirteen days. Rushes took place about the 20th to 25th August, when several flocks passed Isle Ornsay and Lochindaul; again, Sept. 16th to 25th, a flock at Devaar, Portpatrick and Chickens Rock; another rush at Rhu Stoir, Portpatrick and Lochindaul about Oct. 27th. Lapwings travelled in clear or haze, rain, or showers, principally in S. and S.E. winds, on the west coast; but also in N. wind on Aug. 20th, and in W. and S.W. at Rhu Stoir

and Lochindaul on Oct. 23rd and Nov. 1st respectively They appear to travel equally by day and night.

PLOVER.—Three stations give returns of this species, viz., Skerryvore, Rhinns of Islay, and Portpatrick. The earliest date is July 27th, at Portpatrick. The latest date is Oct. 18th, at Skerryvore. The time thus occupied by the migration was fifty-three days. The principal rush took place about the beginning of September, and the favoured station was Portpatrick, where they were always observed prssing inland with a S. to S.W. wind, light or strong breeze, during the daytime. Individuals struck lantern of Rhinns of Islay at night, when wind was N., N.W., or S.E.

OYSTERCATCHER, *Hæmatopus ostralegus*.—At Isle Ornsay on Nov. 18th, one struck at 10.30 p.m.; wind S., light, with haze, and rain.

HERON.—At Dhuheartach on Oct. 8th, one seen at 7 a.m.; calm, fog. At Rhuvaal, Oct. 1st to 30th, "Storks" (afterwards identified as Herons) passed at 9 a.m.; wind variable, light, with showers.

CURLEW.—Island Glass is the most northern station noted for this species; southwards at Dhuheartach, Pladda, Portpatrick and the Mull of Galloway. The earliest date recorded is Aug. 3rd, when eighteen were seen passing Portpatrick at 10 a.m., wind E.S.E., strong breeze, with haze; the latest date is Nov. 20th, when eight were seen passing Island Glass, flying southward, at 2 p.m., wind light S.W., with clear weather Thus the time occupied was one hundred and nine days. A rush took place in August—say 3rd to 26th—and again in November, 15th to 20th. Dhuheartach seemed to be a favourite station. Curlews migrated in all winds, and "boxed the compass" from E.S.E., through S. to S.W., W. and N.N.W.; they passed mostly at night. Two exceptions were at 10 a.m. on Aug. 3rd, and 2 p.m. on Nov. 20th, at Portpatrick and Island Glass respectively.

WHIMBRELS ("Small Curlews"), *Numenius phæopus*.—At Lochindahl on Sept. 20th, seen for several days in flocks; wind S. to W., and rainy.

SANDPIPER (sp.?)—At Rhinns of Islay on Aug. 22nd, one struck at 1 a.m.; wind S.E., light, and haze.

TURNSTONE, *Strepstilas interpres*.—At Rhinns of Islay on Nov. 8th, one struck at night; wind S.W., fresh, and haze.

SNIFE.—At Isle Ornsay, a single Snipe takes position as the northernmost recorded at the lighthouses of the west coast in 1879, as late as Oct. 20th, killed at 10 p.m.; wind N., clear. Thence, records at Skerryvore, Dhuheartach, Rhinns of Islay, Pladda, Mull of Galloway, and Douglas Head. The earliest date, Aug. 20th ("sp. not known"), 4 a.m. at Pladda; wind S.E., strong breeze, and haze.^[16] As a doubt occurs, next earliest date—a long jump later—is Oct. 11th at Rhinns of Islay when one was killed between 12 p.m. and 3 a.m.; wind N., light, with haze. The latest date is Nov. 15th, when, at Rhinns of Islay, sixteen were killed between 6 p.m. and 4 a.m., wind S., breezes and haze; and another record occurs at Pladda. Thus from Oct. 11th to Nov. 15th marks thirty-five days for passage. Snipe are thus seen to make a rapid passage. (Local migrations occur earlier from moors to lowland marshes in Scotland, September always producing home-bred birds in our lowland marshes.) Rush of foreign birds middle of November, 1879, as far as we can judge; but on Oct. 18th "numbers" kept about all night round lantern of Skerryvore; wind S.E., light breeze, fog, and rain. Migration mostly at night, judging from records, but also by day.

^[16] This may have been a Sandpiper.

WOODCOCK.—The northernmost station on west coast in 1879 is Island Glass; thence southwards at Dhuheartach, Rhinns of Islay, Lochindaul, Devaar, Corsewall, Mull of Galloway, and Douglas Head. Small numbers recorded in all cases, except at Lochindaul, when about forty passed during the daytime on Dec. 12th, when the wind was variable, light, and the weather clear. The earliest date was Oct. 11th; one killed at Rhinns of Islay at 10 p.m., wind N., light, clear; the latest Dec. 12th. Too few data to fix rushes, except the one in December, which no doubt was directly caused by the severe frosts then recorded; wind N. and W., except on two occasions, on Oct. 7th at Douglas Head, wind E.; and on Oct. 30th at Mull of Galloway, wind E.; and on two others, viz., variable and light, when (in the hard frost) the forty were seen.

CORN CRAKE, *Crex pratensis*.—Heard for first time at Lochindaul, June 2nd; at Kyleakin, June 9th, wind S.W., clear; and at Skerryvore, June 20th, wind strong S.E., with fog.

WILD GESE (sp.?).—At Monach Island a flock "mixed old and young" passed south at 10 a.m. on Dec. 13th, wind S.S.W., fresh breeze, with haze;

this is the most northerly station noted. Thence southward they were seen at Kyleakin (passing N.W.) at 8 a.m., wind N.W., fresh breeze, and clear weather; also at Sound of Mull, M'Arthur's Head and Devaar. The earliest date is Aug. 17th (passing N.W.), and the latest Dec. 15th (twenty-four passing W., at Sound of Mull). A migratory movement is thus seen to have extended over one hundred and nineteen days. A rush took place on Oct. 20th and 21st, at Sound of Mull and M'Arthur's Head; and again on December 18th, 14th and 15th, at Monach Island, M'Arthur's Head, and Sound of Mull respectively; the direction of their flight was usually W. or N.W. On Dec. 13th a flock passed S.; most passed with a S., S.S.W. or S.W. breeze, in cloudy or clear weather; in two instances with a N.W. wind. They were observed during daylight.

WILD SWAN.—At Devaar on Dec. 11th, six seen at 4 p.m., light airs, with haze; and on Jan. 4th, three were seen flying south, in clear weather, at Rhuvaal lighthouse. We say nothing of the reported occurrence of *Cygnus americanus* on our coasts, but think the record (p. 111) premature.

WILD DUCK.—At Butt of Lewis a Duck was killed on Dec. 1st, at 8 p.m., wind E., light and clear. Others occurred in November and December at Rhuvaal and Devaar. At Rhuvaal, between Nov. 1st and 30th, three or four hundred Ducks passed between 8 a.m. and 3 p.m.; wind variable, light, and showery weather. Most of the migration seems to have been observed through the day. In none of the reports are the species identified, being described as either "Ducks" or "Wild Ducks."

WIDGEON, *Anas penelope*.—At Monach Island on Oct. 27th, a few were seen at 12 noon; wind S.S.E., fresh, and clear.

SHELDRAKE, *Tadorna vulpanser*.—At Pladda, on May 23rd, six (two males and four females) seen at 1 p.m., wind N.E., light, and haze; remained on the island till the middle of June.

EIDER DUCK.—At Rhu Stoir on June 30th, two males and two females seen at 11 a.m.; wind S., fresh breeze, showers. At Dhuheartach on Dec. 3rd, twenty seen at 10 p.m.; wind N.E., light, clear. At Devaar on Dec. 13th, twelve seen at 1 p.m.; wind N.N.W., light breeze, haze.

STORM PETREL.—At Butt of Lewis in July, August, and September, an occasional bird at the glass, always at night, always in hazy or rainy

weather, generally "haze and rain" together; wind in three out of four instances S.W., and once on July 3rd, W.N.W., stormy. At Rhu Stoir a good many were observed "not on passage" between 17th and 20th Aug. between 6 p.m. and 10 p.m., wind N. to E., light, haze and rain; and at Rhinns of Islay on Sept. 17th, one struck at 2 a.m., wind S., light, and haze.

SEA-FOWL.—The following notes upon sea-fowl as observed at several stations—notably at Cape Wrath—will, we think, prove interesting:—At Cape Wrath, Mr. M'Gill reports all the sea-fowl as deserting their breeding-haunts there on Aug. 16th, and both old and young flying westwards. He roughly estimated the numbers which passed within his ken as follows:—300 to 400 Guillemots, 800 to 900 Razorbills, 800 to 900 Puffins, 100 Scarts, and 300 Sea-gulls. This was performed in an east wind, during hazy or rainy weather (see also under Solan Goose), and all left Cape Wrath in one day and about the same time, viz., 10 a.m. The Gulls passed continuously between 9 a.m. and 3 p.m. Notes from other stations can hardly be considered as applying to anything but local daily migration in search of food, as is constantly witnessed at any rock-bird station on the coast; thus, at Rhu Stoir large numbers passed towards S.W., and are specially noted on June 12th to 15th; also at Kyleakin, a large flock passed southward at 2 p.m., wind S.E. When at the Shiant Isles this summer, at the end of June, I witnessed a regular departure of thousands of Puffins to their feeding-grounds in the Little Minch, about 9 to 10 a.m.

SEA GULLS.—At Cape Wrath on Aug. 10th, three hundred passed west between 9 a.m. and 3 p.m.; wind E., haze, and rain. "Sea Gulls bred at the station, and all left on the same day, both young and old. The migration goes all to the west."

KITTIWAKE, *Rissa tridactyla*.—At Skervuile on May 31st, two at noon flying north; calm, clear.

TERN.—At Dhuheartach, six seen on the evening of Aug. 27th, wind S., cloudy; and four at the same station on the morning of Sept. 4th.

BIRDS UNKNOWN.—At Portpatrick on July 12th, a flock of small birds passed overhead at 8.30 a.m., wind S.S.E., fresh, and rainy; and on Oct. 13th a flock passed inland at 4.20 p.m., wind S.W., and rain; on Nov. 13th "a flock of birds not known" passed inland, wind E. by N., and clear. At Pladda on Aug. 20th, one Snipe, species not known, at 4 a.m., wind S.E., strong, and

haze; and another on Nov. 15th. At Skervuile, on Nov. 14th, a number from 10 p.m. to 2 a.m., wind variable, light, and haze; and others on Oct. 7th at Skervuile; "many killed and fell into the sea."

RED-BREASTED MERGANSER.—At Corsewall, on July 20th, a "strange Duck" (since identified as a Red-breasted Merganser) was killed at glass at 2 a.m.; wind E., light, with haze.

SOLAN GOOSE, *Sula bassana*.—At Cape Wrath and other stations upon our coasts, an extraordinary annual migration of Solan Geese is witnessed. Mr. M'Gill has reported to me specially on that of 1879, and kept as accurate a record as possible of the numbers which he estimated to pass westward. He writes as follows:—"The number of Solan Geese that migrate past Cape Wrath is beyond anyone's power to number, but I have come as near to it as possible. The first half of July the flocks were composed of old birds, and the rest of the time they were mixed old and young." Mr. M'Gill then gives the following list of Solan Geese seen passing west on fifteen days, between 14th July and 9th August, which I reproduce here:—

July	14th.	600 to 700.	9 a.m. to 8 p.m.	Wind E.	Clear.
"	15th.	200.	9 a.m. to 8 p.m.	" E.	Haze.
"	16th.	40 to 90.	12 noon to 7 p.m.	" E.	Haze.
"	22nd.	200 to 400.	10 a.m. to 7 p.m.	" N.-N.E.	Fog, rain.
"	24th.	100 to 200.	8 a.m. to 6 p.m.	" S.W.	Clear.
"	26th.	300 to 400.	11 a.m. to 7 p.m.	" S.W.	Haze.
"	28th.	90 to 100.	7 a.m. to 8 p.m.	" S.W.	Haze, rain.
"	31st.	60 to 80.	10 a.m. to 5 p.m.	" S.	Haze, showers.
Aug.	1st.	100 to 300.	8 a.m. to 6 p.m.	" W.	Haze, rain.
"	3rd.	300 to 400.	9 a.m. to 6 p.m.	" E.	Clear.
"	4th.	60 to 70.	10 a.m. to 7 p.m.	" E.	Fog, clear.
"	5th.	20 to 30.	8 a.m. to 5 p.m.	" N.E.	Fog, rain.
"	7th.	40 to 50.	9 a.m. to 6 p.m.	" N.	Fog, rain.
"	8th.	20 to 40.	8 a.m. to 7 p.m.	" N.	Fog, rain.
"	9th.	20.	9 a.m. to 5 p.m.	" N.	Haze.

Thus it will be seen that from 2150 to 3080 were estimated to pass westward within view of Cape Wrath, between July 14th and August 9th, during fifteen days occupied on migration. All which were observed passed

during the day—say between 7 a.m. and 10 p.m.—and apparently in all winds and weather. From the Butt of Lewis Mr. Edgar supplies the general information that the Solan Geese pass the station, arriving first in May, and leaving in the latter end of October, but giving no particulars as to direction of flight. At Mull of Galloway Solan Geese are reported as passing, on July 19th, in flocks. Mr. N. B. Morrison reports as follows:—"Saw them flying past to W.S.W. in wedge-shaped flocks as if on a passage, or changing locality." This was during the forenoon; wind S., light, and haze. "At 1 p.m. it began to rain, and continued for twenty-four hours; rain gauge at 9 a.m. of 20th, 1.72 inches. On 21st, 22nd, 23rd, and 24th, very strong breeze, and showers; wind W.N.W. to S.W." Again at same station (Mull of Galloway) flocks continued passing all day; wind S.E., fresh, fog, and rain, flying in same direction. On 8th and 9th, wind S., gale; 10th, wind W., strong. In a later note Mr. M'Gill believes "that the Solan Geese strike the land first at Cape Wrath," and that the Geese which pass it are from Suliskerry.

On the west coast of Scotland, also, many observers make particular mention of the scarcity of autumnal migrants in 1879. At several principal and important stations this scarcity is of course most remarked upon, such as Butt of Lewis, Monach Island, Island Glass, Skerryvore, and Dhuheartach. During long experience at these and other stations, the several observers do not remember such great scarcity of birds during the autumn migration. From other observers' remarks it would appear that several of the west coast stations are not suitable for observations being made, from their land-locked situations or other local influences, such as Kyleakin, Sound of Mull, Corran Ferry, and others.

Many birds are killed at the lanterns of the more isolated lighthouses and are blown into the sea. Thus, in 1877, at Skerryvore, in the month of October, the number of birds killed was six hundred, chiefly the Common and "Mountain Thrush" (Ring Ouzel), but including also Blackbirds, Snipes, Larks, and one Wild Duck. The observer, Mr. W. Crow, was of opinion that about two hundred more were killed and blown into the sea. They came every night from the 1st to the 6th, about 8 p.m., and went away at daylight. "I would estimate," he says, "the number about the light on each

of the above nights to be about a thousand." The direction of the wind was from S.S.E. to S., with haze; and no migration of birds was observed during the day. On Dhuheartach lighthouse rock, "two Hawks are seen every morning" while the migration lasts, which come to prey upon the small birds resting on the rock. A considerable flight of migrants took place about Oct. 7th, as upon the east coast.

The direction of the flight of migrants appears to be from east to west at the north coast stations, but from N.W. and N.N.W. to S.W. or S.S.W. at the stations farther south; and this is borne out by previous observations in former years by my west coast observers at Tyree and elsewhere. Our observers have not usually reported the direction taken by the migrants, but there are enough data to prove the above lines. It would appear, therefore, that birds when passing from east to west often overshoot the land, and are compelled to turn back upon a new course, according to the direction of the wind. In 1878, Mr. W. Boyd (since deceased) wrote regarding the migration in Mull as follows:—"In the month of October I was fishing on Loch Assapol, near Bunessan. Almost every day I saw flock after flock of little birds—Larks, Buntings, Robins, and even Wrens—flying across the loch. All these birds were steering the same course, having apparently come from the outlying Hebrides, *viá* Tyree, Iona, up the Rose of Mull, and were steering for the mainland. Fresh arrivals of different species of Ducks rested and then passed on. Wild Swans and Geese were seen far up in the air, all taking a bee-line for the south." On another occasion Mr. Boyd visited Tyree in December, 1878, and both he and a companion remarked "the extraordinary scarcity of common birds, and the unusual number of winter visitors." One day every Snipe they put up, instead of flying a bit and settling again, rose high in air, and went off due south-east as far as they could see, right across the sea, to Mull. The remark was then made, "The sooner we go south for powder and provisions the better; we are going to have an arctic winter," which, as is well remembered, was a perfectly correct surmise. It will thus be seen that the flights of wildfowl almost invariably are from N. to S. on both coasts, but that the smaller birds—land birds—as they fly lower, are more influenced by the configuration of the coast-lines, and also, no doubt, by the direction of the wind at the time. Our data are at present too scanty to lay down with precision the minutiae of their lines of flight, but another year's observations will probably greatly assist us. The same rules, as to time of day or night at which birds strike the

lantern, holds upon the west coast which also obtain on the east, and the same remarks as to weather also hold good.

The above remarks upon the direction of the flights upon our west coast may prove of value in a comparison with Mr. Cordeaux's notes upon the direction of the flights at the Galloper Bank. We are aware here that birds pass overland on migration, crossing Scotland between the Firths of Clyde and Forth, as they have been heard on calm nights crying as they passed over from W. to E., or from points N. of W. to points S. of E.; and day-flights have often been observed passing here from N.N.W. towards S.S.E., or from N.W. to S.W. I would instance here Bramblings, *Fringilla montifringilia*, natives of Northern Europe. (See Gray's 'Birds of the West of Scotland,' p. 137. The "column" of Bramblings there described as on migration were not "proceeding in a north-easterly direction," however, as stated by Mr. Gray, but were coming from a north-westerly direction, and were proceeding in a south-easterly direction.)

Transcriber Note

Minor typos were corrected. Several quotation closures were missing and added in the most likely place. Table of Contents added to assist reader.

*** END OF THE PROJECT GUTENBERG EBOOK REPORT ON THE
MIGRATION OF BIRDS IN THE AUTUMN OF 1879. [FIRST REPORT]

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